# Endogenous Plucking Through Networks: The Plucking Paradox*

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**Abstract**

We study an endogenous plucking mechanism in a parsimonious general equilibrium model where firms choose production linkages (network intensity). In expansions, high productivity encourages denser linkages, raising efficiency and output. Yet greater connectedness amplifies adverse TFP shocks, so downturns are disproportionately severe when recessions begin from a highly connected state. The model implies network-size and duration dependence of shock propagation. The calibrated decentralized equilibrium underinvests in network intensity on average relative to the Pareto-efficient allocation because of markups. We characterize an optimal fiscal policy that decentralizes the Pareto-efficient allocation and delivers a Plucking Paradox: the planner prefers higher output in normal times even though it increases exposure to rare, severe disasters.

**Keywords**: Endogenous plucking, production network, duration dependence, optimal fiscal policy, plucking paradox

**JEL codes**: E32, E62, D43, D58

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# 1 Introduction

Business cycles are rarely symmetric fluctuations around a trend. As noted by Friedman (1964), economic activity often resembles a "plucking model": recessions are abrupt, deep deviations below potential, while expansions are gradual returns toward a ceiling. Early formal evidence for such asymmetry in macroeconomic time series is provided by Neftçi (1984). Recently, Hall and Kudlyak (2022a,b) sharpen this view by documenting systematic regularities in recessions and recoveries that are consistent with plucking. Why do long, tranquil expansions so often end in sharp breaks? And why do recoveries display such regular patterns across episodes? Understanding the origins of this asymmetry and historical regularity is key to explaining why fragility builds during periods of prolonged stability.

We provide a parsimonious theory of endogenous plucking based on production networks. In our framework, firms endogenously choose a network intensity, $\alpha$, which governs the strength of input linkages. While higher $\alpha$ boosts efficiency by expanding the use of intermediate inputs, it simultaneously makes aggregate output more sensitive to productivity shocks. Crucially, because $\alpha$ is a sluggish state variable—subject to convex adjustment costs and partial depreciation—the economy cannot instantaneously unwind its network exposure when productivity deteriorates. This "double-edged" nature of networks makes mature booms intrinsically more fragile than young ones.

Our key contribution is twofold. First, we demonstrate that endogenous network accumulation generates nonlinear shock propagation with two distinct forms of state dependence. We prove that the optimal network choice is procyclical: firms gradually accumulate linkages during expansions. At the same time, we show that the output response to a negative shock is strictly increasing in pre-shock network intensity. Together, these results imply: (i) size dependence—the impact of a negative shock scales with the network state—and (ii) duration dependence—recessions are deeper following longer expansions because network intensity has had more time to build up. In equilibrium, the economy "crawls" up an efficiency frontier during expansions but suffers disproportionately severe contractions when productivity turns, reproducing the plucking property endogenously. We validate this mechanism empirically: industries with higher network intensity experience significantly deeper peak-to-trough contractions in employment and payroll, and network intensity itself—proxied by the intermediate-input share—is procyclical, declining during recession episodes exactly as the model predicts.

Our second contribution characterizes the Pareto-efficient allocation and establishes a novel normative result. In the calibrated economy, the retail markup lowers average

network intensity relative to the planner allocation. Correcting it raises output in normal times but also magnifies contractions following adverse productivity shocks. We call this the Plucking Paradox: the efficient economy is more productive on average yet more exposed to severe downturns. The welfare gains from higher normal-time output outweigh the losses from sharper recessions, so efficiency requires accepting rather than eliminating endogenous fragility. We also characterize a fiscal policy that decentralizes this allocation: a constant subsidy on network varieties sold to the retailer that exactly offsets the markup.

In good times, deep input linkages are socially valuable because they improve resource allocation across intermediate varieties and reduce effective production costs. However, like physical capital, these linkages are slow to adjust: once the economy reorganizes around a complex supply chain, this structure cannot be costlessly unwound when conditions deteriorate. Correcting the markup therefore entails a tradeoff in the calibrated economy, shifting it toward a higher-linkage allocation that is more productive but also more pluckable.

**Related literature** This paper is closely related to four strands of literature. First, we contribute to the literature on production networks and macroeconomic fluctuations. Starting with Hulten (1978), a large literature takes the production network as given and studies how input-output linkages shape the propagation and aggregation of shocks (Acemoglu et al., 2012; Carvalho et al., 2020; Baqaee and Farhi, 2019; Acemoglu, Ozdaglar, and Tahbaz-Salehi, 2017).[1] Other work endogenizes production structure itself (Oberfield, 2018; Acemoglu and Azar, 2020; Kopytov et al., 2024; Taschereau-Dumouchel, 2025). Related work studies endogenous risk exposure (Kopytov, Taschereau-Dumouchel, and Xu, 2025) and production scalability, which dampens negative shocks (Kopytov, Taschereau-Dumouchel, and Xu, 2026). In our model, inherited network instead amplifies adverse productivity shocks, generating asymmetry and duration dependence. On the nominal side, Ghassibe (2021) shows how nominal rigidities interact with endogenous link formation to generate state-dependent monetary transmission. Ghassibe and Nakov (2025) instead hold the production network fixed but endogenize firms' repricing decisions, generating pricing cascades whose effects depend on the type of shock. Our mechanism operates on the real side: even with flexible prices, network intensity is predetermined when aggregate productivity shocks arrive, so firms' prior production-structure choices shape the amplification of current shocks. Methodologically, our model is solved globally using

[1] Section 4.6 relates our network multiplier to Hulten's theorem and the nonlinear aggregation results of Baqaee and Farhi (2019).

the repeated transition method developed by Lee (2026b), without relying on the perfect foresight assumption.

Second, we connect to the literature on the plucking model of business cycles (Friedman, 1964; Kim and Nelson, 1999). Kim and Piger (2002) identify a stochastic trend and transitory plucking component by embedding the permanent income hypothesis within a simple growth model. Theoretical accounts of the mechanisms that endogenously generate plucking dynamics are more limited. A notable recent contribution is Dupraz, Nakamura, and Steinsson (2025), who show that a search-and-matching model with downward nominal wage rigidity can generate plucking dynamics in unemployment. We propose a complementary real-side mechanism in which plucking dynamics arise from endogenous production-structure accumulation: expansions gradually build network intensity, which raises efficiency in good times but, being predetermined when productivity shocks arrive, amplifies adverse shocks and generates negative skewness and strong duration dependence.

Third, our paper relates to the literature on the efficiency and resilience of production networks. Ferrari and Pesaresi (2026) study a related productivity–resilience tradeoff in supply chains with search frictions. Their equilibrium features over-specialization because firms fail to internalize how specialization slows supplier replacement after disruptions; the planner therefore prefers less specialization and faster recovery. In our calibrated economy, correcting the retail markup raises average network intensity, so the Pareto-efficient allocation features denser supply chains and greater exposure to aggregate downturns. Capponi, Du, and Stiglitz (2024) analyze whether supply networks are efficiently resilient under perfect foresight. We complement this work by embedding endogenous network formation in a stochastic dynamic environment in which aggregate shocks interact with inherited network intensity. Correcting the markup distortion raises average productivity but also increases endogenous fragility, producing the Plucking Paradox.

Finally, our mechanism relates to work on intermediate inputs and endogenous scalability. Basu (1995) shows that "roundabout" production can magnify the real effects of demand disturbances under imperfect competition even when production exhibits constant returns to scale. Our model likewise maintains constant returns to scale, but endogenizes intermediate-input intensity, which is predetermined when productivity shocks arrive. A related literature studies margins that change returns to scale, including falling IT prices (Lashkari, Bauer, and Boussard, 2024), product standardization (Argente et al., 2025), and input complementarity over the business cycle (Hyun, Kim, and Lee, 2024). Our mechanism is distinct: network formation changes the economy's exposure to productivity

shocks without changing the returns to scale of the production technology.

**Roadmap** The remainder of the paper proceeds as follows. Section 2 introduces the model and equilibrium. Section 3 establishes the monotone buildup of network intensity in expansions and the amplification of negative shocks, delivering size and duration dependence. Section 4 quantifies the mechanism and documents the implied asymmetry in simulated data. Section 5 characterizes the Pareto-efficient allocation and derives the optimal fiscal policy. Section 6 discusses empirical implications and validation. Section 7 concludes.

# 2 Baseline model

This section introduces a parsimonious dynamic stochastic general equilibrium model with continua of symmetric atomistic firms. Network firms endogenously choose their production network intensity, which is the key endogenous state variable.

**Environment** The economy features two unit continua of goods-producing firms: network firms (indexed by $j$) that use a continuum of intermediate inputs whose composition is determined endogenously, and simple firms (indexed by $l$) that use labor only. Both the network sector and the simple sector sell to (i) the network sector as intermediate input for network goods, and (ii) the representative retailer for producing final goods. A representative retailer aggregates network and simple goods into the final consumption good. Throughout, $A$ denotes aggregate TFP, and $w$ denotes the wage. A representative household owns all firms, supplies labor, and consumes the final good.

**Technology** Each network firm produces with one continuum of intermediate inputs. Firm $j$'s network intensity is $\alpha_j \in [0,1]$: a fraction $\alpha_j$ of its intermediate inputs are network goods and the remaining fraction $1-\alpha_j$ are simple goods. We index its network inputs by $i \in [0,\alpha_j]$ and its simple inputs by $k \in [\alpha_j,1]$. The network production function combines these inputs through a Cobb–Douglas aggregator,

$$y_j^N = A \exp\left(\int_0^{\alpha_j} \ln x_{ij}^N \,\mathrm{d}i + \int_{\alpha_j}^1 \ln x_{kj}^S \,\mathrm{d}k\right), \tag{1}$$

where $x_{ij}^N$ and $x_{kj}^S$ are intermediate inputs of network and simple goods, respectively. Simple firms use labor only,

$$y_l^S = A\ell_l. \tag{2}$$

The aggregate TFP process follows $A_{t+1} \sim \Pi_A(\cdot \mid A_t)$, a two-state Markov chain with support $\{G, B\} \subset \mathbb{R}_+$, where $B < 1 < G$.

**Retail aggregation** A representative retailer combines a CES aggregate of network varieties and a Cobb–Douglas aggregate of simple varieties into the final consumption good. Let $X_j^N$ and $X_l^S$ denote the quantities purchased by the retailer from network firm $j$ and simple firm $l$, respectively:

$$Y = \frac{1}{\zeta^\zeta (1-\zeta)^{1-\zeta}} \left[ \left( \int_0^1 (X_j^N)^{\frac{\sigma-1}{\sigma}} \mathrm{d}j \right)^{\frac{\sigma}{\sigma-1}} \right]^\zeta \left[ \exp\left( \int_0^1 \ln(X_l^S)\, \mathrm{d}l \right) \right]^{1-\zeta}. \tag{3}$$

Here $\zeta \in (0,1)$ is the network share in final demand and $\sigma > 1$ is the elasticity of substitution across network varieties. The price of the retailer's final goods is normalized to be the numeraire.

**Market structure** Network firms participate in perfect competition in the intermediate input markets, and monopolistic competition in selling to the retailer. Simple firms operate under perfect competition in both intermediate input and final goods markets. Final-use and intermediate-use markets are segmented, so goods purchased in one market cannot be resold in the other.

**Market clearing** Let $L$ denote aggregate labor demand. Labor, network-goods, and simple-goods market conditions are:

$$\begin{aligned} \int_0^1 \ell_l \, \mathrm{d}l &= L, \\ \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\} \, \mathrm{d}i + X_j^N &= y_j^N, \quad \forall j, \\ \int_0^1 x_{li}^S \mathbf{1}\{i \text{ buys from } l\} \, \mathrm{d}i + X_l^S &= y_l^S, \quad \forall l. \end{aligned} \tag{4}$$

Here $\mathbf{1}\{i \text{ buys from } j\}$ equals one when network firm $i$ buys from network firm $j$, and zero otherwise; the simple-firm indicator is defined analogously.

## 2.1 Recursive competitive equilibrium

We formulate firms' current production decisions and intertemporal network choices jointly. Time is discrete. Let $\alpha_t$ denote the network stock inherited by an individual network firm at the beginning of period $t$, before depreciation, and let $\alpha_t^{\text{agg}}$ denote its aggregate counterpart. The aggregate state is $\mathcal{X}_t = [\alpha_t^{\text{agg}}, A_t]$. At the beginning of the period, a fraction $\delta \in (0,1)$ of inherited links expires, so current production evaluates (1) at $(1-\delta)\alpha_t$. During period $t$, the firm reconfigures its network and chooses the stock $\alpha_{t+1}$, denoted by $\alpha'$ below, that is carried into the next period. The timing is given by Figure 1.

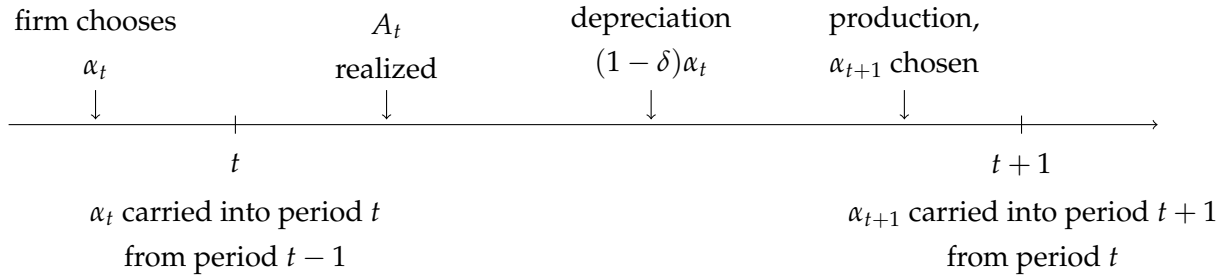



Figure 1: Timing of shocks, production, and network choice

We suppress time subscripts below and write $\mathcal{X} = [\alpha^{\text{agg}}, A]$. The full equilibrium system and its derivation are provided in Appendix H. Let $p_j^{N,\text{final}}$ and $p_l^{S,\text{final}}$ denote the prices at which network firm $j$ and simple firm $l$ sell to the retailer, while $p_j^{N,\text{int}}$ and $p_l^{S,\text{int}}$ denote their prices in the intermediate input markets.

**Retailer's problem** The retailer takes all final-market input prices as given and chooses final output and input quantities. Its problem is

$$\Pi^F(\mathcal{X}) = \max_{Y,\{X_j^N\},\{X_l^S\}} \left\{ Y - \int_0^1 p_j^{N,\text{final}} X_j^N \, \mathrm{d}j - \int_0^1 p_l^{S,\text{final}} X_l^S \, \mathrm{d}l \right\} \quad \text{subject to (3).} \tag{5}$$

**Simple firms' problem** A simple firm $l$ hires labor and sells its output to the retailer and to network firms. It is a price taker and takes $\{p_l^{S,\text{final}}, p_l^{S,\text{int}}, w\}$ as given. Its problem is

$$\Pi_l^S(\mathcal{X}) = \max_{\ell_l, X_l^S, \{x_{li}^S\}} \left\{ p_l^{S,\text{final}} X_l^S + p_l^{S,\text{int}} \int_0^1 x_{li}^S \mathbf{1}\{i \text{ buys from } l\} \, \mathrm{d}i - w\ell_l \right\}$$
$$\text{subject to} \qquad X_l^S + \int_0^1 x_{li}^S \mathbf{1}\{i \text{ buys from } l\} \, \mathrm{d}i \le A\ell_l = y_l^S. \tag{6}$$

The multiplier on the production constraint (6) is denoted by $\lambda_l^S$.

**Decentralized choice of network intensity** A network firm $j$ enters the period with inherited network intensity $\alpha$, of which $(1-\delta)\alpha$ remains productive. It chooses

$$\left\{\alpha',\ p_j^{N,\text{final}},\ \{x_{ij}^N\}_{i\in[0,(1-\delta)\alpha]},\ \{x_{kj}^S\}_{k\in[(1-\delta)\alpha,1]}\right\}.$$

The firm takes as given

$$\left\{\mathcal{X},\ p_j^{N,\text{int}},\ \{p_i^{N,\text{int}}\},\ \{p_k^{S,\text{int}}\},\ \{x_{ji}^N\mathbf{1}\{i \text{ buys from } j\}\},\ X_j^N(\,\cdot\,;\mathcal{X}),\ M(\mathcal{X},\mathcal{X}'),\ \Gamma_{\text{endo}}\right\}.$$

The two price collections are the prices of its network and simple inputs, while the order collection records the intermediate orders placed with firm $j$. The function $X_j^N(\,\cdot\,;\mathcal{X})$ is the demand curve implied by the retailer's problem, $M(\mathcal{X},\mathcal{X}')$ is the stochastic discount factor between periods with states $\mathcal{X}$ and $\mathcal{X}'$, and $\Gamma_{\text{endo}}$ is the firm's perceived law of motion for aggregate network intensity $(\alpha^{\text{agg}})' = \Gamma_{\text{endo}}(\mathcal{X})$.

The firm's operating profit is

$$\begin{aligned}\Pi_j^N\left(\alpha, p_j^{N,\text{final}}, \{x_{ij}^N\}, \{x_{kj}^S\}; \mathcal{X}\right) \equiv\ & p_j^{N,\text{final}} X_j^N(p_j^{N,\text{final}};\mathcal{X}) + p_j^{N,\text{int}} \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i \\ & - \int_0^{(1-\delta)\alpha} p_i^{N,\text{int}} x_{ij}^N\,\mathrm{d}i - \int_{(1-\delta)\alpha}^1 p_k^{S,\text{int}} x_{kj}^S\,\mathrm{d}k. \end{aligned} \tag{7}$$

Following (7), its value satisfies the following Bellman equation

$$J(\alpha;\mathcal{X}) = \max_{\substack{\alpha'\in[0,1),\, p_j^{N,\text{final}},\\ \{x_{ij}^N\},\{x_{kj}^S\}}} \left\{ \Pi_j^N - \Phi(\alpha,\alpha') + \mathbb{E}\left[M(\mathcal{X},\mathcal{X}')J(\alpha';\mathcal{X}')\right] \right\} \tag{8}$$

subject to the sales constraint

$$X_j^N(p_j^{N,\text{final}};\mathcal{X}) + \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i \le y_j^N. \tag{9}$$

Here $y_j^N$ is given by (1), evaluated at network intensity $(1-\delta)\alpha$. The multiplier on the production constraint (9) is denoted by $\lambda_j^N$.

The adjustment cost is

$$\Phi(\alpha,\alpha') = \frac{\mu}{2}\left[\ln\left(\frac{1-(1-\delta)\alpha}{1-\alpha'}\right)\right]^2, \qquad \mu > 0. \tag{10}$$

This form follows directly from combining an so-called urn-ball matching technology (Petrongolo and Pissarides, 2001) with a convex cost of search intensity. Let $\mathcal{U}$ and $\mathcal{V}$ denote the masses on the two sides of the matching market. Given search intensity $s$, the mass of new matches is

$$\mathcal{M} = \mathcal{V}\left[1 - \exp\left(-s\frac{\mathcal{U}}{\mathcal{V}}\right)\right]. \tag{11}$$

This exponential matching structure has early applications in Butters (1977), Hall (1979), and Pissarides (1979). Burdett, Shi, and Wright (2001) provide a decentralized search foundation; Stevens (2007) endogenizes search intensity; and Demir et al. (2024) apply the structure to buyer–supplier formation in a production network.

We assume that producing $s$ requires the convex resource cost $\mu s^2/2$ units of the final good. In our symmetric network economy,

$$\mathcal{U} = \mathcal{V} = 1 - (1-\delta)\alpha, \qquad \mathcal{M} = \alpha' - (1-\delta)\alpha.$$

Eliminating $s$ yields (10).[2] Appendix H states the complete set of static optimality conditions and derives the network firm's envelope and Euler equations.

**Household's problem** The household, with GHH utility, owns all firms. Here $a$ denotes beginning-of-period wealth, and $a'(\mathcal{X}')$ denotes next period's wealth contingent on state $\mathcal{X}'$. Consumption and labor supply are denoted by $c$ and $n$; $\beta \in (0,1)$ is the discount factor, $\rho > 0$ governs the curvature of the GHH composite, $\eta > 0$ scales labor disutility, and $\chi > 0$ is the Frisch elasticity. It solves

$$\begin{aligned} V(a,\mathcal{X}) = \max_{c,n,a'} \quad & \frac{1}{1-\rho}\left(c - \eta\frac{n^{1+1/\chi}}{1+1/\chi}\right)^{1-\rho} + \beta\mathbb{E}\left[V(a',\mathcal{X}')\right] \\ \text{subject to} \quad & c + \mathbb{E}\left[M(\mathcal{X},\mathcal{X}')a'(\mathcal{X}') \mid \mathcal{X}\right] = a + w(\mathcal{X})n, \\ & (\alpha^{\text{agg}})' = \Gamma_{\text{endo}}(\mathcal{X}). \end{aligned} \tag{12}$$

[2] Since $\mathcal{U}/\mathcal{V} = 1$, the matching technology implies $\mathcal{M} = \mathcal{V}(1 - e^{-s})$. Substituting for $\mathcal{M}$ and $\mathcal{V}$ gives $1 - \alpha' = [1 - (1-\delta)\alpha]e^{-s}$, and therefore

$$s = \ln\left(\frac{1-(1-\delta)\alpha}{1-\alpha'}\right).$$

Substitution into the resource cost $\mu s^2/2$ gives (10). The literal matching problem describes new link formation and therefore $s \geq 0$. When the quantitative model permits active link destruction, we interpret $s < 0$ as signed reconfiguration intensity and apply the same quadratic resource cost.

Here $\Gamma_{\text{endo}}$ is the law of motion for the endogenous aggregate state $\alpha^{\text{agg}}$. We assume that the law of motion perceived by the household is consistent with the one perceived by firms.

We can now define the recursive competitive equilibrium.

**Definition 1** (Recursive competitive equilibrium (RCE))**.**
*Given the exogenous TFP transition $A' \sim \Pi_A(\cdot \mid A)$, a recursive competitive equilibrium consists of a price system $q$*

$$q \equiv \left\{ w,\ p_j^{N,final}, p_j^{N,int},\ p_l^{S,final}, p_l^{S,int},\ M(\cdot,\cdot) \;\middle|\; j,l \in [0,1] \right\},$$

*household value and policy functions $(V, g^h)$, firm-side value and policy functions $(J, g^f)$, and a perceived aggregate law of motion $\Gamma_{\text{endo}}$. For every admissible state:*

*(i)* *$(V, g^h)$ solves the household problem (12) given $(q, \Pi_A, \Gamma_{\text{endo}})$;*

*(ii)* *$(J, g^f)$ solves the firm-side problems given $(q, \Pi_A, \Gamma_{\text{endo}})$: the retailer solves (5), every simple firm solves (6), and every network firm solves (8) subject to (9);*

*(iii)* *all markets clear at prices $q$. In particular, (4) holds and household labor supply satisfies $n = L$; and*

*(iv)* *aggregation is consistent: choices are symmetric within firm types,*

$$\alpha_j = \alpha^{\text{agg}}, \qquad \alpha_j' = (\alpha^{\text{agg}})' = \Gamma_{\text{endo}}(\mathcal{X}), \quad \forall j, \forall \mathcal{X}.$$

*Thus the perceived law of motion equals the one generated by individual decisions.*

# 3 A theory of endogenous plucking

This section theoretically investigates the equilibrium properties of the network formation process and its implications for aggregate output fluctuations. The core mechanism rests on a tension between efficiency and stability: complex production networks amplify productivity during booms but create structural fragility that deepens recessions.

To derive analytical closed-form results, we restrict attention to a partial equilibrium setting where the stochastic discount factor is fixed.[3] Our goal is to formally prove that this economy exhibits duration dependence: the longer a boom lasts, the more severe the subsequent recession will be—the hallmark of the plucking model.

[3]Later in the quantitative analysis, we show that the endogenous stochastic discount factor modestly dampens aggregate volatility, while the qualitative theoretical predictions remain unchanged.

## 3.1 Network dynamics during booms

We begin by characterizing how the economy's network structure evolves during a period of sustained productivity growth. Because network adjustments are subject to convex costs, the optimal network intensity $\alpha$ is not a static choice but a state variable that accumulates over time.

**Definition 2** (Equilibrium policy path)**.**
*Let $f(\alpha; \mathcal{X})$ denote the optimal policy function solving the firm's problem. We define the iterated policy function $g^{(n)}(\alpha; G)$ as the network choice after $n$ consecutive periods of state $A = G$, starting from state $\alpha$:*

$$g^{(1)}(\alpha; G) := f(\alpha; G, \alpha) \tag{13}$$

$$g^{(n)}(\alpha; G) := f(g^{(n-1)}(\alpha; G); G, g^{(n-1)}(\alpha; G)) \quad \forall n \in \{2, 3, \dots\} \tag{14}$$

*This sequence characterizes the evolution of network intensity during a prolonged boom.*

The following proposition establishes that network formation is self-reinforcing. A higher existing stock of network capital reduces the marginal cost of maintaining or expanding connections, creating momentum in network accumulation.

**Proposition 1** (Weak monotonicity in network size)**.**
*The policy function $g^{(1)}(\alpha; G)$ is weakly increasing in $\alpha$. That is, $\frac{\partial g^{(1)}}{\partial \alpha} \geq 0$.*

*Proof.* In the partial-equilibrium problem, the firm chooses $\alpha' \in \mathcal{A} \subset [0, 1]$ to maximize

$$F(\alpha', \alpha; G) \equiv \Pi(\alpha'; G) - \Phi(\alpha, \alpha'),$$

where $\mathcal{A}$ does not depend on $\alpha$. Assume $F(\cdot, \alpha; G)$ is strictly concave in $\alpha'$ so that the optimizer is unique and characterized by the first-order condition

$$F_{\alpha'}(\alpha', \alpha; G) = 0.$$

Since $\Pi(\alpha'; G)$ does not depend on $\alpha$, we have

$$F_{\alpha'\alpha}(\alpha', \alpha; G) = -\Phi_{\alpha'\alpha}(\alpha, \alpha').$$

Using the adjustment cost in (10), a direct calculation gives

$$\Phi_{\alpha'\alpha}(\alpha, \alpha') = -\frac{\mu(1-\delta)}{(1-\alpha')[1-(1-\delta)\alpha]} < 0,$$

and therefore $F_{\alpha'\alpha}(\alpha', \alpha; G) > 0$.

Let $\alpha_2 > \alpha_1$. Consider the function $\psi(\alpha') \equiv F_{\alpha'}(\alpha', \alpha_2; G) - F_{\alpha'}(\alpha', \alpha_1; G)$. Because $F_{\alpha'\alpha} \geq 0$, we have $\psi(\alpha') \geq 0$ for all $\alpha'$, i.e., the marginal benefit of increasing $\alpha'$ is (weakly) higher when the inherited state $\alpha$ is larger. Since $F(\cdot, \alpha; G)$ is strictly concave, $F_{\alpha'}(\cdot, \alpha; G)$ is strictly decreasing in $\alpha'$. Hence the unique zero of $F_{\alpha'}(\cdot, \alpha_2; G)$ must occur at a (weakly) higher $\alpha'$ than the unique zero of $F_{\alpha'}(\cdot, \alpha_1; G)$, implying $g^{(1)}(\alpha_2; G) \geq g^{(1)}(\alpha_1; G)$. ■

Strict concavity of $F(\cdot, \alpha; G)$ is used only to guarantee a unique optimizer characterized by the first-order condition; it is not needed for monotonicity itself. Because $\Pi(\alpha'; G)$ is independent of $\alpha$ and $\Phi_{\alpha'\alpha} < 0$, the objective $F$ exhibits increasing differences in $(\alpha', \alpha)$, so by Topkis's monotonicity theorem (Topkis, 1978) the set of maximizers is nondecreasing in $\alpha$ in the strong set order, and the conclusion of Proposition 1 holds for its maximal selection without any concavity assumption. In our quantitative solution the policy function is single-valued, so the two formulations coincide.

**Corollary 1** (Iterated weak monotonicity)**.**
*The n-th iterated policy $g^{(n)}(\alpha; G)$ is weakly monotone increasing in $\alpha$.*

*Proof.* This follows by induction from Proposition 1. Since $g^{(1)}$ is non-decreasing, and composition of non-decreasing functions preserves monotonicity, $g^{(n)}$ remains non-decreasing. ■

Proposition 1 implies that history matters: the network choice today depends on the network choice yesterday. The next logical step is to determine the direction of this accumulation during a boom. We show that if firms find it optimal to increase network intensity at the start of a boom, they will continue to do so as the boom persists.

**Proposition 2** (Monotone preservation property)**.**
*If the initial network adjustment is positive, i.e., $g^{(1)}(\alpha; G) \geq \alpha$, then subsequent adjustments maintain this trajectory: $g^{(2)}(\alpha; G) \geq g^{(1)}(\alpha; G)$.*

*Proof.* This property relies on the weak monotonicity established in Proposition 1. Let $g(\alpha)$ denote the policy function $g^{(1)}(\alpha; G)$. We know from Proposition 1 that $g(\alpha)$ is non-decreasing in $\alpha$. Assume the initial condition holds: $g(\alpha) \geq \alpha$. Because $g(\cdot)$ is non-decreasing, we can apply the function $g$ to both sides of this inequality without changing the sign:

$$g(g(\alpha)) \geq g(\alpha) \tag{15}$$

By definition, $g^{(2)}(\alpha; G) \equiv g(g(\alpha))$ and $g^{(1)}(\alpha; G) \equiv g(\alpha)$. Therefore, the inequality simplifies to:

$$g^{(2)}(\alpha; G) \geq g^{(1)}(\alpha; G) \tag{16}$$

Intuitively, if the firm wants to grow the network today (investment is positive), the higher resulting network state tomorrow will motivate them to grow it even further (or at least maintain it) due to the complementarity of the network stock. ■

**Corollary 2** (Iterated monotonicity preservation)**.**
*If* $g^{(1)}(\alpha; G) \geq \alpha$*, then* $g^{(n)}(\alpha; G) \geq g^{(n-1)}(\alpha; G)$ *for all* $n \in \{2, 3, \dots\}$.

*Proof.* Let $T$ be the operator such that $\alpha_{t+1} = T(\alpha_t) = g^{(1)}(\alpha_t; G)$. If $T(\alpha) \geq \alpha$, applying the monotone operator $T$ to both sides (by Proposition 1) yields $T(T(\alpha)) \geq T(\alpha)$, which is equivalent to $g^{(2)}(\alpha; G) \geq g^{(1)}(\alpha; G)$. By induction, $\alpha_n$ is a non-decreasing sequence converging to the steady state associated with $A = G$. ■

Taken together, these results paint a picture of gradual network buildup. During a boom, firms do not immediately jump to the high-network steady state due to adjustment costs. Instead, $\alpha$ rises monotonically over time. Consequently, a "mature" boom (large $n$) is characterized by a strictly higher network intensity than a "young" boom (small $n$).

## 3.2 Endogenous fragility and plucking

Having established that network intensity accumulates during booms, we now examine how this structure performs when the economy is hit by a negative shock. We show that the network acts as a double-edged sword: the same leverage that amplifies output in good times amplifies the contraction in bad times.

**Proposition 3** (Output monotonicity in network)**.**
*Conditional on the bad aggregate state* $A = B$*, aggregate output* $Y(\alpha, B)$ *is strictly decreasing in* $\alpha$*.*

*Proof.* From the static equilibrium, output satisfies

$$Y(\alpha, A) = A^{\frac{\zeta}{1-(1-\delta)\alpha}+1} L.$$

Under GHH preferences, labor supply satisfies $n(\alpha, A) = \big(w(\alpha, A)/\eta\big)^{\chi}$ and equals to labor demand $L$ in equilibrium. and the static wage is

$$w(\alpha, A) = \left(\frac{\sigma - 1}{\sigma}\right)^{\zeta} A^{\frac{\zeta}{1-(1-\delta)\alpha}+1}.$$

Therefore $L \propto A^{\chi\left(\frac{\zeta}{1-(1-\delta)\alpha}+1\right)}$, and hence

$$Y(\alpha, A) \propto A^{(1+\chi)\left(\frac{\zeta}{1-(1-\delta)\alpha}+1\right)}.$$

Taking logs and differentiating yields

$$\frac{\partial \ln Y}{\partial \alpha} = (1+\chi)\frac{\zeta(1-\delta)}{\big(1-(1-\delta)\alpha\big)^2}\ln A.$$

Since $(1+\chi)\frac{\zeta(1-\delta)}{(1-(1-\delta)\alpha)^2} > 0$, the sign is determined by $\ln A$. Because $B < 1$ implies $\ln B < 0$, we obtain $\partial Y(\alpha, B)/\partial \alpha < 0$. ■

This result identifies the source of endogenous fragility. Define the network multiplier $\Gamma(\alpha) \equiv \frac{\zeta}{1-(1-\delta)\alpha}$, so that equilibrium output satisfies $Y \propto A^{(1+\chi)(\Gamma(\alpha)+1)}$: the elasticity of output with respect to TFP is increasing in the network state. A high $\alpha$ implies a high elasticity. When TFP is low ($A = B < 1$), a high elasticity magnifies the decline.[4]

Finally, we connect the network accumulation dynamics (Corollary 2) with this fragility result (Proposition 3) to formally derive the plucking property.

**Corollary 3** (Size dependence)**.**
*$Y(g^{(n)}(\alpha; G), B)$ weakly decreases in the initial state $\alpha$.*

*Proof.* From Corollary 1, $g^{(n)}(\alpha; G)$ increases in $\alpha$. From Proposition 3, $Y(\cdot, B)$ decreases in the network argument. The composition of a decreasing function and an increasing function is decreasing. ■

**Corollary 4** (Duration dependence)**.**
*If the economy starts with $g^{(1)}(\alpha; G) \geq \alpha$, then $Y(g^{(n)}(\alpha; G), B)$ weakly decreases in $n$ for all $n \in \{2, 3, \dots\}$.*

*Proof.* By Corollary 2, the sequence of network states $\{\alpha_n\}_{n=1}^{\infty}$ generated by a sequence of positive shocks is non-decreasing in $n$. Let $\alpha_n = g^{(n)}(\alpha; G)$. Since $\alpha_n \geq \alpha_{n-1}$ and $\frac{\partial Y(\alpha,B)}{\partial \alpha} < 0$ (Proposition 3), it follows that:

$$Y(\alpha_n, B) \leq Y(\alpha_{n-1}, B) \tag{17}$$

[4]It is worth noting that the result in Proposition 3 is not an artifact of normalizing $A$ around one. Let $\bar{A}$ denote the reference productivity level that pins the stationary allocation (e.g., the unconditional mean or the deterministic steady state), and define $\tilde{A} \equiv A/\bar{A}$. Then the comparative statics can be written as

$$\frac{\partial \ln Y}{\partial \alpha} = \frac{\zeta(1-\delta)}{\big(1-(1-\delta)\alpha\big)^2}\ln \tilde{A} \quad (\times \text{a positive labor term, discussed below}),$$

so networks raise output when productivity is above its reference level ($\tilde{A} > 1$) and lower output when productivity is below it ($\tilde{A} < 1$). The mechanism is therefore structural: network intensity increases the elasticity of output with respect to TFP deviations from normal times, which is precisely why inherited network exposure turns expansions into fragile peaks.

This establishes the theoretical basis for endogenous plucking: the severity of the recession (output level in state $B$) is monotonically increasing in the duration of the preceding boom. ■

In summary, the model generates business cycle asymmetry not through asymmetric shocks, but through the endogenous evolution of the state variable. A long boom allows the economy to build an interconnected, efficient network structure. When the inevitable downturn arrives, the inherited high-network state amplifies its effect, so the same adverse shock produces a deeper recession after a long boom than after a short one.

# 4 Quantitative equilibrium analysis

## 4.1 Disciplining the baseline model

We calibrate the baseline model to match annual U.S. moments describing production structure and aggregate fluctuations. Exogenous TFP follows a two-state Markov chain with "good" and "bad" levels, whose transition matrix is

$$\Pi_A \equiv \begin{bmatrix} \Pr(G \mid G) & \Pr(B \mid G) \\ \Pr(G \mid B) & \Pr(B \mid B) \end{bmatrix} = \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix}. \tag{18}$$

Rows index the current state $A_t$ and columns index the next state $A_{t+1}$, both ordered as $(G, B)$. The diagonal probabilities imply an expected duration of $1/(1-0.75) = 4$ years in each regime. The parameters $\{\eta, \mu, \zeta, B, G\}$ are jointly disciplined.

The quantitative analysis proceeds in three steps. First, given parameters, we solve for the recursive equilibrium policy functions for network intensity, prices and output. Second, we simulate long artificial histories under the calibrated Markov process to compute ergodic moments. Lastly, we compare simulated moments to their data counterparts and update parameters until they match.

We target five moments: (i) mean labor supply; (ii) the mean intermediate-input share; (iii) the standard deviation of HP-filtered log output; (iv) mean final-demand-weighted upstreamness, a measure of the average number of production stages in the economy (Antràs et al., 2012); and (v) the skewness of output growth.[5] We normalize mean labor supply to one third of available time. The intermediate-input-share target is a rounded

[5]The intermediate-input share is the ratio of total intermediate expenditure to gross output and, in the model, equals $\zeta(\sigma-1)/\{\sigma[1-(1-\delta)\alpha_t+\zeta]-\zeta\}$. Final-demand-weighted upstreamness equals $1-\zeta+\frac{\zeta[\sigma(2-\zeta)-1]}{\sigma-\zeta}\frac{\sigma-(1-\delta)\alpha_t}{\sigma[1-(1-\delta)\alpha_t]}$. Appendix A derives both mappings.

moment of aggregate materials expenditures relative to shipments in the 1958–2018 NBER–CES Manufacturing Industry Database (Becker, Gray, and Marvakov, 2021). Output volatility and growth skewness are computed from annual U.S. constant-dollar GDP per capita over 1960–2024 from the World Development Indicators, accessed through FRED[6]. We compute the upstreamness target from the BEA input–output tables using the measure developed by Antràs et al. (2012).

The five moments jointly discipline $\{\eta, \mu, \zeta, B, G\}$. Mean labor supply is particularly informative about the labor-disutility parameter $\eta$. The mean intermediate-input share and upstreamness jointly inform the network-goods share $\zeta$ and the ergodic level of network intensity. Output volatility and growth skewness jointly discipline the adjustment-cost parameter $\mu$ and the location and dispersion of the two-state productivity process: $\mu$ governs the endogenous response of network intensity, while $B$ and $G$ determine the productivity levels. All five free parameters are calibrated jointly rather than assigned to moments one by one.

Table 1: Value of parameters and exogenous variables

| Symbol | Description | Value | Reference |
|---|---|---|---|
| **Preferences and technology** | | | |
| $\rho$ | Risk aversion | 1.000 | log utility |
| $\beta$ | Discount factor | 0.960 | annual model frequency |
| $\delta$ | Depreciation rate of network | 0.100 | annual breaking rate $= 10\%$ |
| $\sigma$ | Elasticity of substitution | 6.000 | markup $\approx 20\%$ |
| $\eta$ | Disutility of labor | 2.568 | joint calibration (Table 2) |
| $\chi$ | Frisch elasticity of labor | 1.000 | Keane and Rogerson (2012) |
| $\zeta$ | Share of network goods | 0.856 | joint calibration (Table 2) |
| $\mu$ | Adjustment cost parameter | 0.007 | joint calibration (Table 2) |
| **Aggregate fluctuation** | | | |
| $G$ | TFP value of "good time" | 1.004 | joint calibration (Table 2) |
| $B$ | TFP value of "bad time" | 0.996 | joint calibration (Table 2) |
| $\Pi_A$ | TFP transition matrix | $\begin{bmatrix} 0.750 & 0.250 \\ 0.250 & 0.750 \end{bmatrix}$ | $\mathbb{E}[\text{regime duration}] = 4$ years |

*Notes:* Annual calibration. The Frisch elasticity is set to one. For each candidate $\{\mu, \zeta, B, G\}$, $\eta$ is chosen to normalize labor supply to one third at the reference productivity level. The four remaining parameters are jointly calibrated to the corresponding moments in Table 2.

The quantitative mechanism is driven by the predetermined nature of network intensity: $\alpha_{t+1}$ is chosen in period $t$. A high network choice in a good state can therefore

[6]Federal Reserve Economic Data, series `NYGDPPCAPKDUSA`

Table 2: Calibration moments

| Moment | Data moment | Model moment |
|---|---|---|
| Mean labor supply | 0.333 | 0.334 |
| Mean intermediate-input share | 0.550 | 0.550 |
| Standard deviation of HP-filtered log output | 0.013 | 0.013 |
| Mean upstreamness | 1.780 | 1.780 |
| Skewness of output growth, 1960–2024 | -0.553 | -0.553 |

*Notes:* Mean labor supply is normalized to one third of available time. The intermediate-input-share target is a rounded moment from the 1958–2018 NBER–CES Manufacturing Industry Database (Becker, Gray, and Marvakov, 2021). Output volatility is computed from annual U.S. constant-dollar GDP per capita over 1960–2024 from the World Development Indicators, accessed through FRED (series `NYGDPPCAPKDUSA`). The upstreamness target is computed from the BEA input–output tables using the measure of Antràs et al. (2012). Model moments use the 9500 annual observations in the formal calibration sample.

carry into a subsequent bad state and amplify the contraction. Adjustment costs shape the persistence and gradual evolution of this inherited exposure.

## 4.2 Solution method

The key contribution of this paper lies in the analysis of state-dependent equilibrium dynamics. Local linearization techniques are not suitable for this purpose: by construction, they approximate the policy function around a single steady state and cannot capture the sign-switching behavior that is central to our mechanism—the marginal effect of network intensity on output is positive when $A > 1$ and negative when $A < 1$. Global solution methods are therefore essential.

We adopt the Repeated Transition Method (RTM) developed by Lee (2026b), which globally solves the recursive competitive equilibrium without imposing perfect foresight or relying on local approximations. The core idea is to exploit the ergodicity of the equilibrium allocations: if the economy is simulated for a sufficiently long horizon under the true stochastic process, nearly all relevant combinations of endogenous states (network intensity $\alpha$) and exogenous states (productivity $A$) will be realized along the equilibrium path.

This property enables an accurate computation of conditional expectations directly from the simulated history. To evaluate $\mathbb{E}[V(\alpha', A') \mid \alpha, A]$, we identify all periods along the simulated path where the economy visits approximately the same endogenous state $\alpha$, but experiences different subsequent realizations of the exogenous shock $A'$. By averaging continuation values across these periods, we obtain a nonparametric estimate of the

conditional expectation that respects the global structure of the value function—including the state-dependent asymmetries that are central to our results.

### 4.3 Benchmark results

We simulated the calibrated model for 10,000 periods using the RTM. Figure 2 summarizes the dynamics of the benchmark model. The displayed window reports a portion of the full simulated path. The shaded areas represent recession periods ($A = B$).

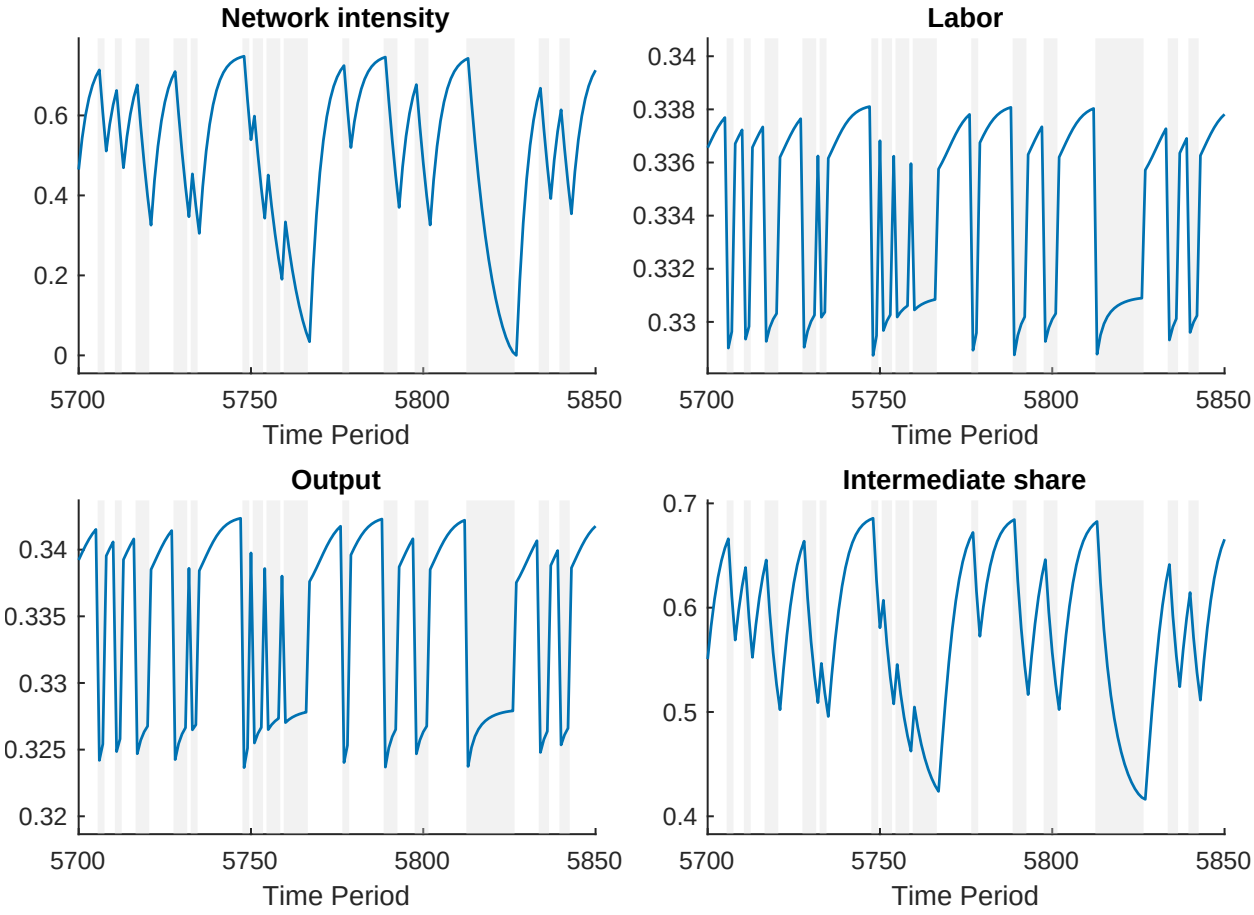



Figure 2: Simulated sequence for $\alpha_t$, $N_t$, $Y_t$, and intermediate share

Notes: Simulated 10,000 periods under the baseline calibration. Panels show the joint dynamics of network intensity $\alpha_t$, labor $N_t$, output $Y_t$, and the intermediate-input share. Shaded areas indicate recession regimes ($A = B$). The figure illustrates the gradual buildup of $\alpha_t$ in good times and the sharp contraction in $Y_t$ and $N_t$ when a bad TFP draw hits a high-network state.

The simulated paths display asymmetry: network intensity $\alpha$ builds up slowly in good times due to adjustment costs, but when a bad productivity draw arrives, the $\alpha$ inherited from previous good periods amplifies the drop, causing a sharp plunge in output $Y$, and labor $N$. After the initial plunge, there are three stages of slow recovery. First, during the

subsequent bad periods after the initial shock, the network remains a "liability" for firms, and they optimally choose to shrink it, but due to adjustment cost this process is gradual. Second, when recession is over, and productivity recovers, the network already shrunk to a low level which limits the network amplification effect in good times, leading to a slow recovery of output. Third, the gradual rebuilding of the network in high productivity times due to adjustment costs then further prolongs the recovery phase.

This contrasts with standard RBC capital: capital accumulation in booms amplifies output in good times and tends to buffer bad shocks, whereas network participation behaves like capital in booms but amplifies downturns in recessions.

To further quantify the regime dependence of the model, we follow Krusell and Smith (1998) and estimate the following linear projections separately for the good and bad regimes:

$$\begin{aligned} \alpha_{t+1} &= \beta_1(A_t)\alpha_t + \beta_2(A_t) \\ \log(N_t) &= \beta_3(A_t)\alpha_t + \beta_4(A_t) \\ \log(Y_t) &= \beta_5(A_t)\alpha_t + \beta_6(A_t) \end{aligned}$$

This regime dependence is summarized in Table 3.

Table 3: Regime-specific linear projections

| | Bad state ($A = B$) | | | Good state ($A = G$) | | |
|---|---|---|---|---|---|---|
| Variable | $\alpha_{t+1}$ | $\log N_t$ | $\log Y_t$ | $\alpha_{t+1}$ | $\log N_t$ | $\log Y_t$ |
| $\alpha_t$ | 0.867 | -0.008 | -0.015 | 0.750 | 0.010 | 0.020 |
| Constant | -0.017 | -1.105 | -1.113 | 0.192 | -1.093 | -1.089 |
| $R^2$ | 0.999 | 0.908 | 0.908 | 1.000 | 0.914 | 0.914 |
| Observations | 4784 | 4784 | 4784 | 4716 | 4716 | 4716 |

*Notes:* OLS projections use the post-burn-in benchmark simulation and are estimated separately in bad ($A = B$) and good ($A = G$) states. The first dependent variable is next period's network intensity; labor and output are in logs. The opposite signs of the $\alpha_t$ coefficients for $\log N_t$ and $\log Y_t$ reflect the model's state-dependent network amplification.

The slope coefficients for $\log(N_t)$ and $\log(Y_t)$ are negative in the bad state and positive in the good state, while the network policy remains increasing in inherited network intensity in both regimes. In contrast to Krusell and Smith (1998), where the law of motion for aggregate capital features similar coefficients across regimes, here the allocation rules for labor and output exhibit slopes of opposite signs across regimes. This sign flip is a struc-

tural implication of the model's elasticity channel: network intensity raises the elasticity of output and wages with respect to TFP deviations, so inherited $\alpha$ amplifies expansions when $A$ is above its reference level and amplifies contractions when $A$ is below it. For $\log(N_t)$, the channel runs through equilibrium wages: $\partial \log w / \partial \alpha$ has the same sign as $\log A$. With GHH utility, labor supply depends on the real wage but has no wealth effect, so $N_t$ inherits the same sign flip—higher $\alpha$ raises labor in good states but lowers it in bad states.

## 4.4 Size dependence and duration dependence

**Network-size dependence.** We first isolate the size dependence in the dynamics implied by Corollary 3: the impact of an adverse shock scales with inherited network intensity. We compute generalized impulse response functions (GIRFs) as follows. First, from periods immediately following $A = G$, we select the low, median, and high values of inherited network intensity. Second, for each selected state, we generate 5,000 paired stochastic paths. Within each pair, one path begins in $G$, while the other begins in $B$ on impact; thereafter, both paths use common Markov innovations and evolve according to the same transition matrix. We use the regime-specific policy functions recovered from the ergodic path to propagate network intensity. The GIRF is the average output difference between the $B$-initialized and $G$-initialized paths, expressed as a percentage of ergodic mean output.

Figure 3 reports the resulting responses. The impact decline increases from 3.15% at the low network state to 5.69% at the high network state: the same negative shock produces a substantially deeper contraction when it hits a denser inherited network. The vertical gap between the two conditional saddle paths at a given $\alpha$ (Section 4.5) is exactly this impact effect, so the diverging saddles explain why the GIRFs fan out with the initial network state.

**Duration dependence.** Figure 4 provides a time-domain illustration of Corollary 4, holding the initial network state fixed.[7] Both paths start from the same network state, remain in $G$ for either eight or two periods, and then face the same switch from $G$ to $B$. Output falls more sharply after the longer good spell: the impact decline is 5.23 percentage points, compared with 3.90 percentage points after the shorter spell. Because the exogenous shock is identical across the two paths, the difference reflects the additional network intensity accumulated during the longer boom.

[7] An interactive visualization of this exercise is available at https://ye-sun-econ.github.io/endogenous_plucking_demo/.

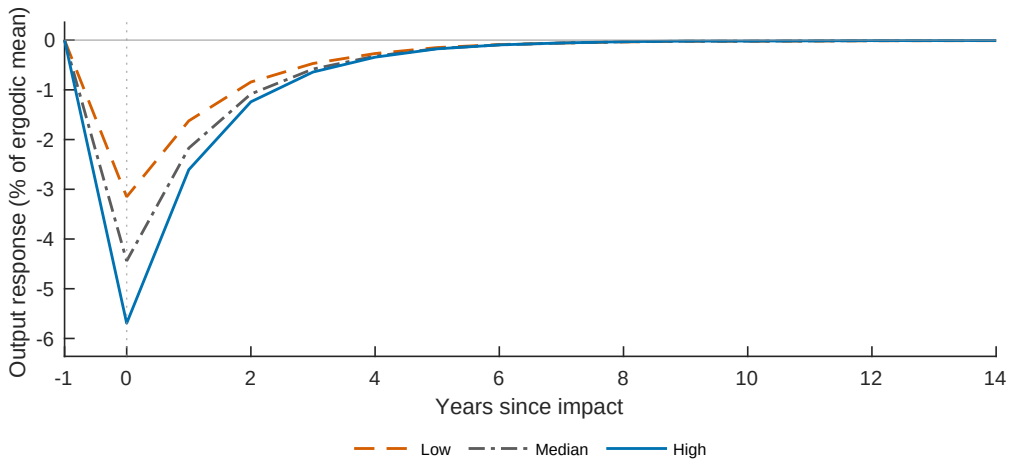



Figure 3: Network-size dependence: generalized impulse response functions by inherited network intensity

Notes: GIRFs are computed from 5,000 paired stochastic paths at low, median, and high inherited network intensities. Within each pair, one path begins in $G$, the other begins in $B$, and both paths use common future Markov innovations. The plotted response is the average output difference between the $B$-initialized and $G$-initialized paths, expressed as a percentage of ergodic mean output.

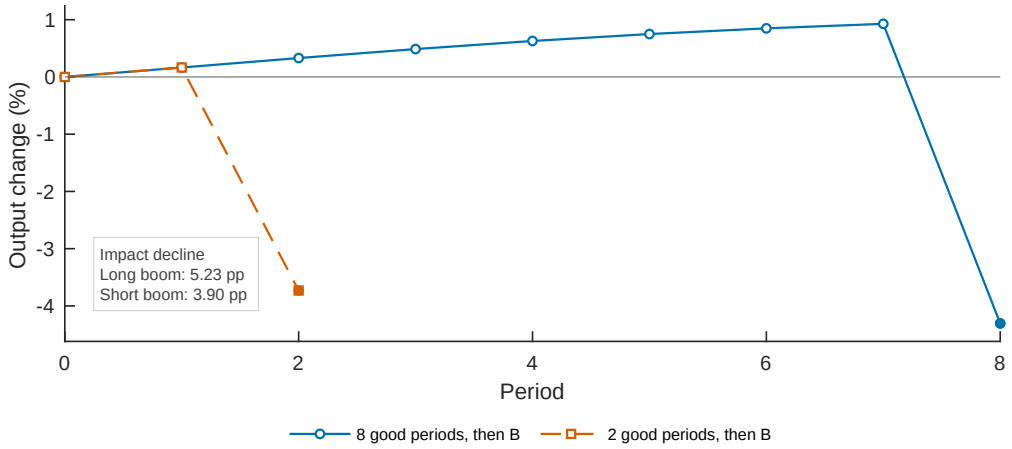



Figure 4: Corollary 4: duration dependence

Notes: Both conditional paths start from the same network state. We interpolate the equilibrium policy and output mappings from the benchmark simulation, hold productivity at $G$ for eight or two periods, and then switch it to $B$. Output is expressed as a percentage change from the common initial output level, and the two paths are aligned at that initial period. The impact decline is the change from the final $G$ observation to the first $B$ observation.

## 4.5 Endogenous plucking

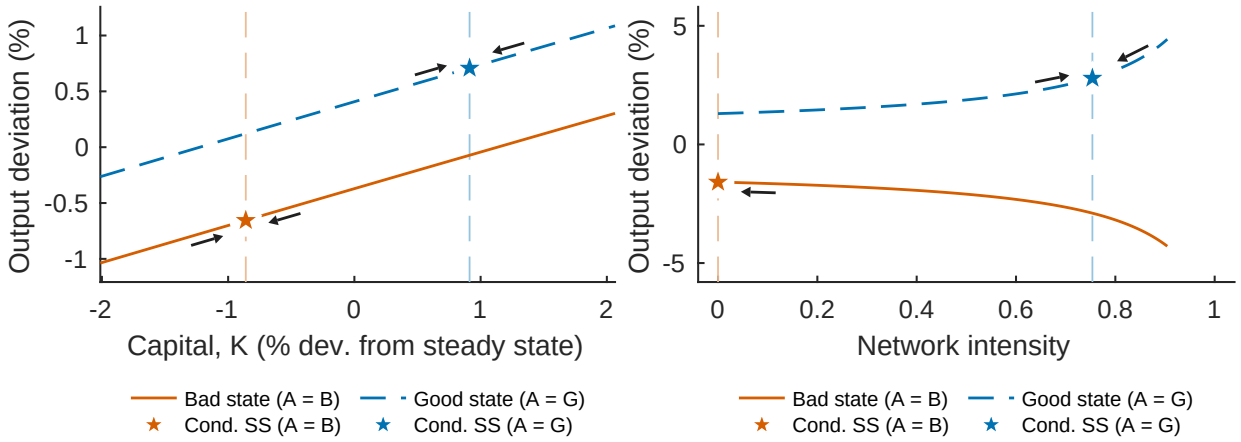
The theoretical results in Section 3 establish that network intensity builds monotonically during booms (Corollary 2) and amplifies contractions (Proposition 3). These two proper-

ties have a single geometric representation through conditional saddle paths (Lee, 2026a), which also deliver a model-implied trend–plucking decomposition in the spirit of Kim and Piger (2002).

**Conditional saddle paths.** Formally, fix $A \in \{G, B\}$ and an initial network state $\alpha_0$. Holding the realized productivity state at $A$, repeatedly apply the equilibrium law of motion

$$\alpha_{t+1} = \Gamma_{\text{endo}}([\alpha_t, A]), \qquad Y_t = Y(\alpha_t, A).$$

The resulting orbit $\{(\alpha_t, Y_t)\}_{t\geq 0}$ is the conditional saddle path associated with $A$. Importantly, $\Gamma_{\text{endo}}$ is the policy from the full stochastic recursive competitive equilibrium: firms continue to evaluate future regime changes according to $\Pi_A$; only the realized sequence plotted here is held at $A$. In the calibrated equilibrium, the orbit converges to a regime-specific fixed point $(\alpha_A^{\text{cs}}, Y_A^{\text{cs}})$, which we refer to as the regime-specific attractor. Thus, along a realized path, the economy follows one conditional saddle while the regime persists and moves to the other when the regime switches.




(a) Canonical RBC: $(K, Y)$ (b) Network model: $(\alpha, Y)$

Figure 5: Conditional saddle paths: RBC benchmark versus endogenous networks.

Notes: Each panel plots conditional saddle paths—the equilibrium transition loci under frozen regimes $A \in \{G, B\}$—computed from the globally solved model. Stars and the corresponding vertical dashed guides mark the regime-specific attractors; arrows indicate motion toward them. Panel (a): canonical RBC in $(K, Y)$ space; both loci slope upward, differing mainly by a level shift (sign-preserving state dependence). Panel (b): network economy in $(\alpha, Y)$ space; the $G$ locus slopes upward while the $B$ locus slopes downward (sign-reversing state dependence). The vertical distance between the two loci at any $\alpha$ equals the impact effect of a regime switch $\Delta Y(\alpha)$ defined in (19). This distance widens with $\alpha$, so longer booms—which push $\alpha$ rightward along the $G$ path—produce deeper contractions when the regime switches to $B$.

**RBC benchmark: sign-preserving state dependence.** Panel (a) of Figure 5 plots the conditional saddle paths for the canonical RBC economy, providing a benchmark for comparison with our network economy.[8] In contrast to the network economy, the RBC attractor changes its nature across regimes: it is a ceiling in $G$ (output lies below and climbs toward it) but a floor in $B$ (output lies above and descends toward it). Hence the one-sided recovery from below that defines Friedman's plucking is absent in the RBC benchmark. This difference reflects the geometry of the conditional saddle paths. In the RBC benchmark, both conditional saddle paths slope upward in $(K, Y)$ space. Inherited capital is productive regardless of the regime, so a regime switch generates a vertical displacement without reversing the qualitative relationship between the slow-moving state and output. The two conditional saddles are approximately parallel, differing mainly in their vertical position. State dependence in the RBC model is sign-preserving: a bad shock reduces output by more when capital is higher, but the direction of the effect—capital supports output—is the same in both regimes.

**Network economy: sign-reversing state dependence.** Panel (b) of Figure 5 plots the network economy in $(\alpha, Y)$ space. The geometry is fundamentally different. Under $G$, the conditional saddle slopes upward—higher network intensity raises output during expansions. Under $B$, it slopes downward—inherited network intensity now amplifies contractions. This slope reversal means that the sign of the relationship between the inherited state and output flips across regimes: network intensity is an asset in good times but a liability in bad times.

The upward slope of the $G$ locus reflects the positive effect of network intensity on output in good times, while the rightward arrows depict the monotone network buildup established in Corollary 2: during a sustained expansion, firms gradually accumulate linkages, moving the economy rightward along the conditional saddle toward the high-$\alpha$ conditional steady state $\alpha_G^{cs}$. The downward slope of the $B$ locus is Proposition 3, which proved $\partial Y(\alpha, B)/\partial \alpha < 0$: conditional on bad productivity, output is strictly decreasing in inherited network exposure. The leftward arrows show network intensity declining toward $\alpha_B^{cs}$ when the bad state persists. The conditional saddle diagram thus provides a compact visual representation of the theoretical mechanism that Section 3 established algebraically.

**Plucking as across-saddle geometry.** Because $\alpha$ is predetermined, a regime switch from $G$ to $B$ at inherited state $\alpha$ corresponds to a vertical jump from the $G$ saddle to the $B$ saddle

[8] The canonical RBC model setup for Panel (a) is available in Appendix F.

at the same $\alpha$. The vertical distance between the two conditional saddle paths at a given $\alpha$,

$$\Delta Y(\alpha) \;:=\; Y(\alpha, G) - Y(\alpha, B), \tag{19}$$

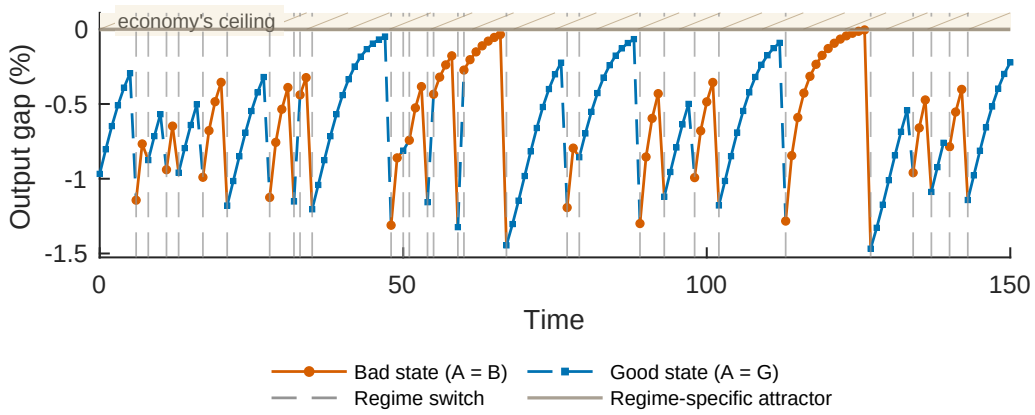
is exactly the magnitude of the impact effect of the regime switch. Because the two saddle paths diverge—one slopes up, the other down—$\Delta Y(\alpha)$ is strictly increasing in $\alpha$. This is the geometric content of size dependence (Corollary 3): the impact of a negative shock scales with the pre-shock network state. It is also what the GIRFs in Figure 3 measure: at a given initial $\alpha$, the GIRF equals $-\Delta Y(\alpha)$ on impact, and the diverging saddles explain why the GIRFs fan out with the initial network state. Duration dependence (Corollary 4) follows immediately, since a longer boom moves $\alpha$ further rightward along the $G$ saddle toward $\alpha_G^{\text{cs}}$, increasing the vertical gap at the point of regime switch.

The distance between the two conditional steady states, $\alpha_G^{\text{cs}} - \alpha_B^{\text{cs}}$, measures the range of plucking exposure: it is the maximal rightward drift that $\alpha$ can undergo during a prolonged expansion before the next contraction. The wider this range, the more scope the economy has to build up fragility during long booms. In the calibrated model, this range is substantial (Figure 5, Panel (b)), consistent with the large output drops observed after extended expansions in the simulations of Section 4.4.

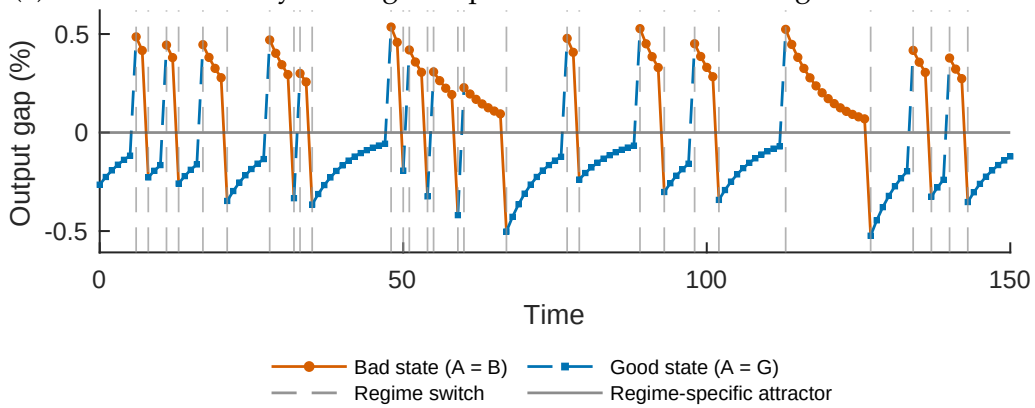
**A model-implied trend–plucking decomposition.** Kim and Piger (2002) decompose output into a stochastic trend and a transitory plucking component. Our model delivers a structural counterpart of this decomposition. The regime-specific attractor is naturally interpreted as Friedman's "ceiling": the economy converges toward that ceiling when the regime persists, and moves to the other ceiling when the regime switches. Denote the log deviation of output from the current regime's ceiling by

$$\hat{y}_t = \log(Y_t) - \log(Y_{A_t}^{\text{CS}}).$$

Along the calibrated equilibrium path, network intensity approaches the $G$-state attractor from below and the $B$-state attractor from above. Combined with the positive slope of the $G$ locus and the negative slope of the $B$ locus, this implies that $\hat{y}_t$ is weakly negative: output lies below the current regime-specific ceiling and converges toward it as long as the regime persists. Figure 6 plots $\hat{y}_t$ along a simulated path and shows this one-sided recovery; the RBC benchmark, by contrast, does not exhibit one-sided recovery from below because its bad-state attractor acts as a floor. The one-sided $\hat{y}_t$ is the model-implied transitory plucking component, and the regime-specific attractor plays the role of the stochastic trend in Kim and Piger (2002).




(a) Network economy: the regime-specific attractor is a ceiling in both $G$ and $B$.




(b) RBC benchmark: the regime-specific attractor is a ceiling in $G$ but a floor in $B$.

Figure 6: Plucking: recovery toward a regime-specific attractor

Notes: This figure plots the network economy (top panel) and the RBC benchmark (bottom panel) under the same realized productivity-state sequence and over the same time window. The vertical axis is the log deviation of output from the current regime's attractor. Vertical dashed lines mark regime switches. The horizontal line at zero denotes the economy's regime-specific ceiling for the network economy. In the network economy, the path illustrates one-sided recovery toward that ceiling within each regime rather than symmetric fluctuations around a fixed trend. The lower panel plots $\hat{y}_t$ in the RBC benchmark: $\hat{y}_t$ is negative under $G$ (the attractor acts as a ceiling) but positive under $B$ (the attractor acts as a floor), so the attractor's nature switches across regimes and the economy does not exhibit one-sided recovery from below.

**Predictability asymmetry.** We ask whether the simulated equilibrium also reproduces the standard predictability asymmetry. Figure 7 applies the turning-point algorithm of Dupraz, Nakamura, and Steinsson (2025) to HP-filtered simulated log output. Following Kim and Nelson (1999) and Kim and Piger (2002), we interpret plucking as an asymmetry in the transitory component of output around a stochastic trend; the HP filter removes the trend, and the attractor decomposition above provides the model-implied alternative. Applying the same turning-point test directly to $\hat{y}_t$ yields the same asymmetry (Appendix Figure D.4). The amplitude of a contraction strongly predicts the subsequent recovery, whereas a recovery has little predictive power for the next contraction. The canonical

RBC benchmark, subject to the same productivity process, exhibits neither relationship. Importantly, this predictability asymmetry is not among the moments targeted in the calibration.

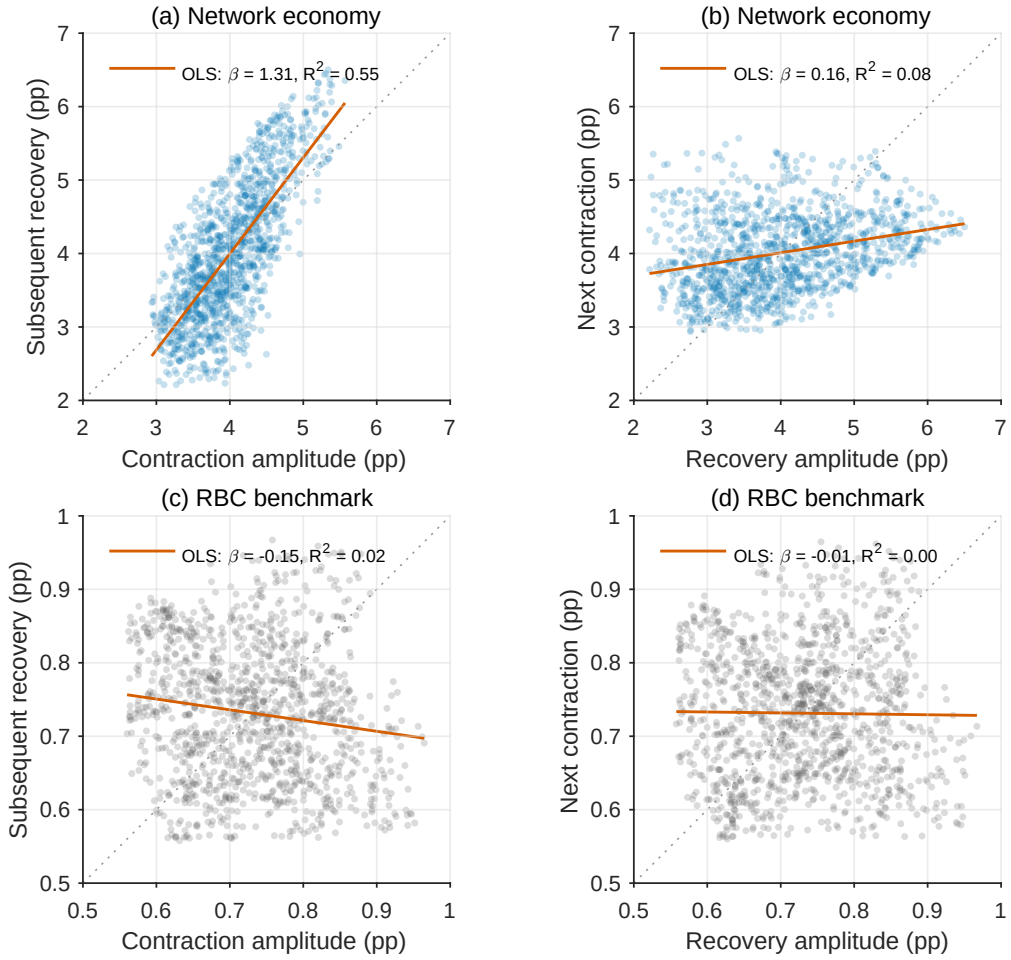



Figure 7: Predictability asymmetry: network economy and RBC benchmark

Notes: We apply the turning-point algorithm of Dupraz, Nakamura, and Steinsson (2025) to minus the cyclical component of simulated log output, obtained with an HP smoothing parameter of 6.25. Turning points must be separated by at least 1.5 standard deviations of the resulting series. Amplitudes are reported as 100 times log changes. Panels (a)–(b) report the network economy, and Panels (c)–(d) report a canonical RBC economy under the same calibrated productivity process. Within each row, the left panel plots each contraction against its subsequent recovery; the right panel plots each recovery against the next contraction. The RBC benchmark uses the same discount factor and depreciation rate as the network economy, a capital share of 0.33, and fixed labor. Orange lines are OLS fits, and gray dotted lines are 45-degree lines. In the network economy, the slopes are 1.31 and 0.16, with $R^2$ values of 0.55 and 0.08. In the RBC benchmark, the corresponding slopes are $-0.15$ and $-0.01$, with $R^2$ values of 0.02 and 0.00. Each row uses 1,204 and 1,203 paired episodes from the post-burn-in simulation. Axis scales differ across rows.

## 4.6 Relation to Hulten (1978) and beyond

We relate our mechanism to the literature of production networks and macroeconomic fluctuations. In an economy with $N$ sectors indexed by $i$, suppose that each sector is subject to a sector-specific TFP shock $\varepsilon_i$. Under standard assumptions, Hulten (1978) shows that aggregate output responds to sectoral productivity shocks according to

$$\log(\text{GDP}) = \sum_{i=1}^{N} \lambda_i \varepsilon_i,$$

where $\lambda_i$ is the Domar weight of sector $i$. These weights are determined by the input-output structure of the economy. If $A$ is the input-output matrix and $d$ denotes sectoral Cobb-Douglas shares in household utility, then $\lambda = (I - A)^{-1} d$. This result is further generalized by Baqaee and Farhi (2019).

Following the logic in Acemoglu et al. (2012), the variance of aggregate output can be linked to two network objects: average network participation and the dispersion of network centrality. To see this, define the operators "$\mathbb{E}$" and "Var" as the finite-sample mean and variance over $i \in \{1, 2, \ldots, N\}$. Assuming that sectoral shocks are independent and identically distributed with variance $\sigma_\varepsilon^2$, aggregate volatility is

$$\sigma_{\text{GDP}}^2 = \sum_{i=1}^{N} \lambda_i^2 \sigma_\varepsilon^2 = N \sigma_\varepsilon^2 \mathbb{E}[\lambda_i^2] = N \sigma_\varepsilon^2 \left( \mathbb{E}[\lambda_i]^2 + \text{Var}(\lambda_i) \right).$$

The Domar weight is the ratio of sectoral sales to GDP, so

$$\begin{aligned} \mathbb{E}[\lambda_i] &= \frac{1}{N} \sum_{i=1}^{N} \lambda_i = \frac{1}{N} \sum_{i=1}^{N} \frac{\text{sales}_i}{\text{GDP}} \\ &= \frac{\text{Gross Output}}{N \cdot \text{GDP}} = \frac{1}{N} \cdot \frac{1}{1 - \text{Total Intermediate Share}}. \end{aligned}$$

Since Bonacich centrality$_i$ is defined as $N\lambda_i$,

$$\text{Var}(\lambda_i) = \frac{\text{Var}(\text{Bonacich centrality}_i)}{N^2}.$$

Substituting these two expressions into aggregate volatility gives

$$\sigma^2_{\text{GDP}} = \frac{\sigma^2_\varepsilon}{N}\left[\underbrace{\left(\frac{1}{1-\text{Total Intermediate Share}}\right)^2}_{\text{network participation}} + \underbrace{\text{Var}(\text{Bonacich centrality}_i)}_{\text{network topology}}\right].$$

This decomposition highlights two channels through which production networks amplify aggregate fluctuations: (i) network participation, or how much firms rely on supply chains for production, and (ii) network topology, or how linkages are organized across sectors, especially the dispersion of centralities.

Our model holds network topology fixed and makes network participation time-varying and endogenous. The network-participation channel is procyclical: firms build network intensity during booms as network inputs amplify productivity gains, yet the same inherited intensity amplifies downturns when productivity is low. This makes the Domar weights procyclical, rendering aggregate volatility and fragility endogenous.

This decomposition also clarifies the role of our representative-network simplification. Even in a model with heterogeneous firms and an arbitrary input-output structure, the network-participation channel would still be fully captured by the total intermediate share, which is a sufficient statistic for log GDP volatility if the network topology channel is held fixed. In the model, firms share a common network intensity $\alpha$, so the topology term is shut down: centralities do not vary across firms and $\text{Var}(\text{Bonacich centrality}_i) = 0$. The single aggregate state $\alpha$ is therefore a sufficient statistic for log GDP volatility.

# 5 The plucking paradox and the optimal fiscal policy

We have established that endogenous network formation generates plucking and business cycle asymmetry. A natural question follows: is this asymmetry efficient? Does the decentralized economy generate "too much" or "too little" fragility?

In this section, we solve the social planner's problem and compare its allocation with the decentralized equilibrium. The retail markup on network varieties is the primitive distortion. Under the calibrated parameters, removing this markup raises average network intensity and consumption, but it also makes output growth more volatile and more negatively skewed.

## 5.1 Planner's problem: Pareto efficiency

The social planner directly chooses all quantities, including final demand for each network variety. Prices therefore do not enter the planner's problem. Before imposing symmetry, the state contains the entire profile $\{\alpha_j\}_{j\in[0,1]}$, and the recursive problem is

$$V^{SP}(\{\alpha_j\}, A) = \max_{\substack{c,n,Y,\{\alpha'_j,\ell_j,X_j^N,X_j^S\}_{j\in[0,1]},\\ \{x_{ij}^N,x_{kj}^S\}_{i,k,j\in[0,1]}}} \left\{u(c,n) + \beta\mathbb{E}\left[V^{SP}\left(\{\alpha'_j\}, A'\right)\middle| A\right]\right\} \tag{20}$$

$$\text{subject to} \qquad c = Y - \int_0^1 \Phi(\alpha_j, \alpha'_j)\,\mathrm{d}j, \tag{21}$$

$$Y = \frac{1}{\zeta^\zeta(1-\zeta)^{1-\zeta}}\left[\left(\int_0^1 (X_j^N)^{\frac{\sigma-1}{\sigma}}\,\mathrm{d}j\right)^{\frac{\sigma}{\sigma-1}}\right]^\zeta \left[\exp\left(\int_0^1 \ln X_j^S\,\mathrm{d}j\right)\right]^{1-\zeta}, \tag{22}$$

$$y_j^N = A\exp\left(\int_0^{(1-\delta)\alpha_j} \ln x_{ij}^N\,\mathrm{d}i + \int_{(1-\delta)\alpha_j}^1 \ln x_{kj}^S\,\mathrm{d}k\right), \qquad j\in[0,1], \tag{23}$$

$$y_j^S = A\ell_j, \qquad n = \int_0^1 \ell_j\,\mathrm{d}j, \tag{24}$$

$$y_j^N = X_j^N + \int_0^1 x_{ji}^N \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i, \qquad j\in[0,1],$$

$$y_j^S = X_j^S + \int_0^1 x_{ji}^S \mathbf{1}\{i \text{ buys from } j\}\,\mathrm{d}i, \qquad j\in[0,1]. \tag{25}$$

The feasible set requires $0 \le \alpha'_j < 1$ for every $j$. Appendix H.2 derives the planner's first-order conditions and then imposes the symmetrical firm assumption.

## 5.2 The markup wedge

Network firms are monopolistic competitors when selling varieties to the retailer. Their decentralized final-market price therefore satisfies $p_j^{N,\text{final}} = [\sigma/(\sigma-1)]\lambda_j^N$, where $\lambda_j^N$ is marginal cost. The planner instead chooses $X_j^N$ directly, so its corresponding shadow-price condition contains no markup. This product-market markup is the primitive wedge between the two allocations.

The same wedge also affects the choice of next period's network intensity. To see it directly, compare how the firm and the planner value the same marginal increase in $\alpha'$. Holding fixed the candidate values of $(\alpha, \alpha', \alpha'')$, future output $Y'$, and the stochastic

discount factor $M(\mathcal{X}, \mathcal{X}')$, their Euler equations are:

$$\text{CE:} \qquad \Phi_2(\alpha, \alpha') = \mathbb{E}\left[ M(\mathcal{X}, \mathcal{X}') \left\{ \frac{\sigma - 1}{\sigma} \zeta Y' \ln A' \frac{1-\delta}{[1-(1-\delta)\alpha']^2} - \Phi_1(\alpha', \alpha'') \right\} \right],$$

$$\text{SP:} \qquad \Phi_2(\alpha, \alpha') = \mathbb{E}\left[ M(\mathcal{X}, \mathcal{X}') \left\{ \zeta Y' \ln A' \frac{1-\delta}{[1-(1-\delta)\alpha']^2} - \Phi_1(\alpha', \alpha'') \right\} \right].$$

The two conditions differ only by the constant markup factor $(\sigma - 1)/\sigma$ multiplying the decentralized gross marginal return. Thus the dynamic wedge is simply the intertemporal implication of the product-market markup.

At the baseline value $\sigma = 6$, the firm internalizes five sixths of the social gross return to network adjustment. The markup therefore acts as a dynamic brake: it weakens network buildup when $A = G$ and network unwinding when $A = B$. Conversely, correcting the markup generates state-dependent adjustment even though the corrective rate itself is constant, because the gross network return changes sign across regimes. This Euler comparison does not imply a pointwise ordering of the two policies. In the calibrated stationary allocation, however, the planner chooses a higher network intensity on average. The same markup also distorts the intratemporal allocation. Conditional on the same $(\alpha, A)$, GHH labor supply satisfies

$$n^{SP}(\alpha, A) = \left( \frac{\sigma}{\sigma - 1} \right)^{\zeta\chi} n^{CE}(\alpha, A). \tag{26}$$

Thus the intratemporal labor distortion and the intertemporal network distortion are two implications of the same retail markup, rather than two distinct market failures. As $\sigma \to \infty$, the markup disappears and the decentralized conditions converge to the planner's conditions.

## 5.3 The optimal fiscal policy

Because the distortion originates in retail pricing, it can be corrected directly at that margin. Let $\tau^R$ denote an ad valorem subsidy on a network firm's sales to the retailer: the retailer pays $p_j^{N,\text{final}} X_j^N$, while the firm receives $(1 + \tau^R) p_j^{N,\text{final}} X_j^N$.

**Proposition 4** (Fiscal implementation)**.**
*The Pareto-efficient allocation is decentralized by a constant subsidy on network varieties sold to*

*the retailer satisfying*

$$1 + \tau^R = \frac{\sigma}{\sigma - 1}, \qquad \textit{or equivalently} \qquad \tau^R = \frac{1}{\sigma - 1}, \tag{27}$$

*financed by a lump-sum tax.*

With the subsidy, the network firm's pricing condition becomes

$$p_j^{N,\text{final}} = \frac{\sigma}{\sigma - 1} \frac{\lambda_j^N}{1 + \tau^R}.$$

Equation (27) therefore sets the price paid by the retailer equal to marginal cost. Intermediate markets are already competitive, so this correction aligns the entire static price system, the household labor condition, and the network firm's Euler equation with their planner counterparts. No separate wage, network-profit, or adjustment-cost subsidy is required. The gross corrective factor is constant—it does not vary with the aggregate state or network intensity—and converges to one as $\sigma \to \infty$. Lump-sum financing makes the subsidy and tax pure transfers in the aggregate resource constraint.

## 5.4 The plucking paradox

Correcting the markup raises average consumption and welfare, but it also makes consumption more volatile and makes contractions following a switch from $G$ to $B$ larger. We call this combination the Plucking Paradox. The planner maximizes the same expected GHH utility, with the same curvature parameter $\rho$, as the representative household in CE. The additional volatility therefore cannot be attributed to a different attitude toward risk; it is an endogenous consequence of correcting the markup.

**A richer but more fragile allocation.** Table 4 reports the central comparison. Mean network intensity rises from 0.4337 in CE to 0.5194 under the planner, while mean consumption rises from 0.3331 to 0.3904, an increase of 17.2 percent. Consumption volatility, measured by $\text{sd}(\log C)$, rises from 0.0200 to 0.0226, a 13.3 percent increase. Because mean labor also rises from 0.3335 to 0.3900, the increase in mean consumption does not translate one-for-one into welfare. The ergodic consumption-equivalent welfare gain is 1.45 percent.[9]

[9]The consumption equivalent is the permanent proportional increase in CE consumption, holding the CE labor path fixed, that equates ergodic expected GHH utility to that under the planner allocation.

Table 4: Ergodic levels and fragility in CE and the planner allocation

| Model | Ergodic $C$ | Ergodic $\alpha$ | $\mathrm{sd}(\log C)$ | Obs. |
|---|---|---|---|---|
| CE | 0.3331<br>(0.3251, 0.3416) | 0.4337<br>(0.0450, 0.7299) | 0.0200 | 9500 |
| Planner | 0.3904<br>(0.3789, 0.4024) | 0.5194<br>(0.0801, 0.8169) | 0.0226 | 9500 |

*Notes:* The first two columns report ergodic means, with the 5th and 95th percentiles in parentheses. The third column reports the standard deviation of log consumption over the simulated ergodic distribution. CE and the planner allocation use the same calibrated parameters and the same productivity sequence.

**Where the higher mean and volatility come from.** The two outcomes arise through different margins. Up to a state-independent constant, aggregate output satisfies

$$\log Y_t = (1+\chi)\left[1 + \frac{\zeta}{1-(1-\delta)\alpha_t}\right]\log A_t.$$

The term in brackets is the elasticity of output with respect to productivity, and it increases with $\alpha_t$. A denser network therefore raises output when $A_t = G > 1$ and lowers it when $A_t = B < 1$. Because log productivity is nearly centered around zero, these opposing effects largely cancel in mean log output but widen the gap between good-state and bad-state output.

To measure the two effects separately, we construct an accounting allocation period by period. We retain the common productivity sequence and the CE network choices $(\alpha_t^{CE}, \alpha_{t+1}^{CE})$. At each date, we recompute labor and output from the planner's intratemporal conditions evaluated at $(A_t, \alpha_t^{CE})$, evaluate adjustment costs as $\Phi(\alpha_t^{CE}, \alpha_{t+1}^{CE})$, and set consumption equal to output minus that cost. Thus,

$$\widetilde{N}_t = n^{SP}(\alpha_t^{CE}, A_t), \qquad \widetilde{C}_t = Y^{SP}(\alpha_t^{CE}, A_t, \widetilde{N}_t) - \Phi(\alpha_t^{CE}, \alpha_{t+1}^{CE}).$$

We do not recompute expectations, value functions, or network choices, and we do not solve the RTM again. This object is therefore an accounting decomposition, not a third equilibrium. Replacing the CE within-period allocation with the planner's mapping while holding the CE network path fixed raises mean consumption from 0.3331 to 0.3902, but leaves $\mathrm{sd}(\log C)$ at 0.0200. Allowing the planner to choose its own network path then barely changes mean consumption, from 0.3902 to 0.3904, while raising consumption volatility to 0.0226.[10] Thus, in this accounting sense, nearly all of the increase in mean

[10]The conclusion does not depend on using the CE network path as the intermediate case. Evaluating the planner network path under the CE within-period mapping instead gives $\mathbb{E}[C] = 0.3333$ and

consumption is an intratemporal efficiency gain, whereas nearly all of the additional volatility is generated by the induced intertemporal network policy.

**Duration dependent SPP-CE difference.** The planner's network policy makes short-run log output growth both more volatile and more negatively skewed. Using the same productivity sequence, the standard deviation of output growth rises from 0.0205 to 0.0234, while its skewness falls from $-0.551$ to $-0.789$. Figure 8 examines both when the larger contractions occur and how frequently large contractions arise. Panel (a) groups each $G$-to-$B$ transition according to the length of the preceding good-state spell. Panel (b) plots the empirical survival function: for each contraction size, it reports the fraction of transitions whose contraction is at least that large, without imposing a parametric tail distribution. Along the calibrated path, consumption rises within each regime spell, so the final $G$ observation is the episode peak and the initial $B$ observation is its trough. The one-period transition drop therefore coincides with the episode's peak-to-trough decline.

Across all $G$-to-$B$ transitions, the mean log-consumption contraction is 4.28 log points in CE and 4.98 log points under the planner. This average masks a pronounced duration gradient. Following one period of $G$, the corresponding contractions are 3.64 and 3.93 log points, a planner–CE difference of 0.28 log point. After seven or more good periods, they reach 5.31 and 6.41 log points, and the difference widens to 1.11 log points. Longer booms therefore increase fragility in both allocations, but more strongly under the planner because the two economies accumulate increasingly different network exposures. Panel (b) shows that the planner survival curve lies weakly above the CE curve at every observed threshold: each of the 1,205 common episodes has a larger contraction under the planner. For a contraction of 5.5 log points, 5.1 percent of CE contractions exceed that threshold ($\sim$ 95% of the CE contraction is smaller than that), compared with 36.2 percent under the planner (only $\sim$ 64% of the planner contraction is smaller than that), suggesting that the planner allocation has a substantially fatter left tail, not just on a larger mean contraction. The planner's allocation therefore exhibits both more frequent and more severe disasters.

Taken together, the accounting and event-level comparisons sharpen the paradox. The intratemporal correction raises mean consumption, while the induced intertemporal network dynamics raise volatility and place substantially more probability on large contractions, especially after long booms. Both are consequences of correcting the same markup.

$\mathrm{sd}(\log C) = 0.0225$: the network change again moves volatility almost to its planner value while leaving mean consumption nearly unchanged.

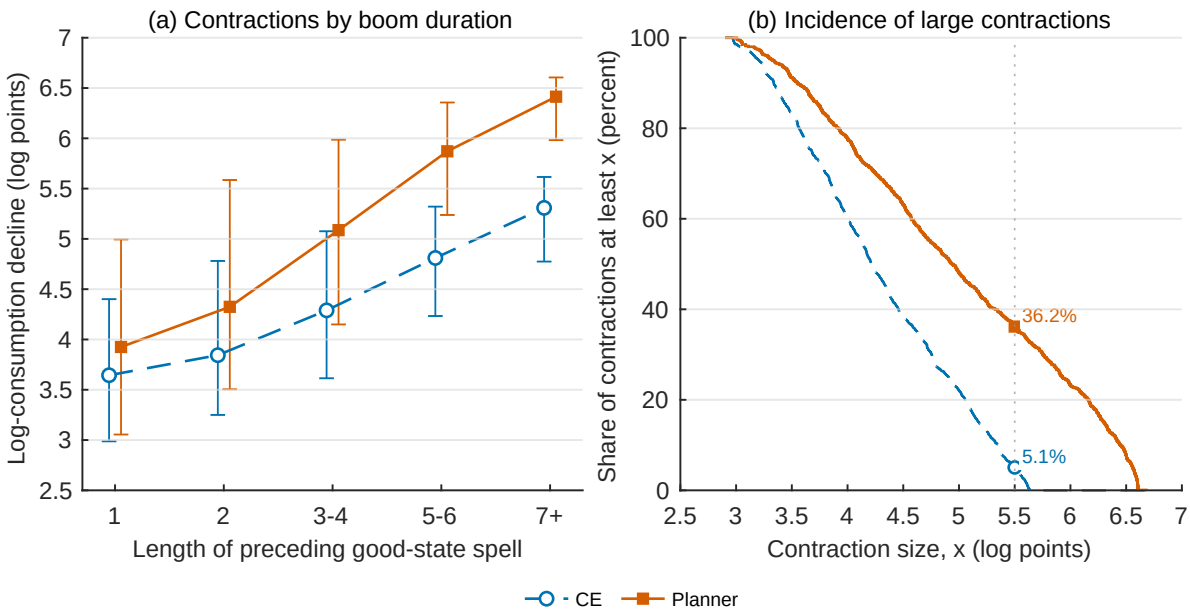



Figure 8: Boom duration and the distribution of consumption contractions

Notes: At each $G$-to-$B$ transition, the contraction for allocation $X \in \{\text{CE}, \text{Planner}\}$ is $100[\log C^X_{t-1} - \log C^X_t]$. Panel (a) reports its mean by the number of consecutive good-state periods ending at $t-1$. Each transition enters once: for example, $BGGGGGB$ contributes one observation with duration five, rather than five observations with durations one through five. The five bins contain 312, 230, 295, 154, and 214 common transitions, respectively. Markers are within-bin means; whiskers span the within-bin 5th to 95th percentiles and therefore describe cross-episode dispersion, not confidence intervals. Panel (b) plots the empirical survival function: at each threshold, it reports the percentage of contractions at least that large. The markers highlight a common threshold of 5.5 log points. Both allocations use the same calibrated parameters, post-burn-in sample, and productivity sequence. Positive values denote contractions.

**The "apologetic" planner.** Greater impact fragility does not imply weaker subsequent adjustment. After consumption falls, the planner has a steeper within-regime mean-reversion slope than the decentralized economy, relative to each allocation's own regime-specific ceiling. Let $\widehat{c}^X_t = \log C^X_t - \log C^{\text{cs},X}_{A_t}$ denote consumption relative to allocation $X$'s regime-specific ceiling ($X \in \{\text{CE}, \text{Planner}\}$), let $P_X$ indicate the planner allocation, and let $G_t$ indicate the good state. On observations with $A_t = A_{t-1}$, we estimate

$$\Delta \widehat{c}^X_t = a + a_P P_X + a_G G_t + a_{PG} P_X G_t + b \widehat{c}^X_{t-1} + d P_X \widehat{c}^X_{t-1} + \varepsilon^X_t.$$

The CE mean-reversion slope is $b = -0.186$, while the planner slope is $b + d = -0.238$. The interaction estimate is $d = -0.0526$, with a paired Newey–West standard error of 0.0006 and a 95 percent confidence interval of $[-0.0538, -0.0513]$. The implied persistence of the consumption gap is therefore 0.814 in CE and 0.762 under the planner; the corresponding half-life of deviations falls from 3.38 to 2.55 periods, or by 24.5%. We call this pattern "apologetic": the planner builds a denser network in good times, experiences

a larger contraction relative to its own preceding level when the bad state arrives, and then moves consumption toward its new ceiling more strongly. This faster mean reversion does not drive the average welfare gain, which comes primarily from the planner's intratemporal allocation.

# 6 Empirical validation

## 6.1 Revisiting evidence for plucking

The plucking view originates with Friedman (1964) and has been tested in a variety of time-series settings. Early formal evidence for asymmetric business cycle dynamics is provided by Neftçi (1984). Subsequent empirical work uses unobserved-components models with Markov switching on U.S. macro series (Kim and Nelson, 1999). Cross-country evidence in Hartley (2021) suggests that plucking-style asymmetry is broadly present across countries, and particularly pronounced in advanced economies. More recent work by Dupraz, Nakamura, and Steinsson (2025) documents the plucking properties in U.S. monthly unemployment data and identifies peaks and troughs that closely align with the NBER chronology.

A related literature asks whether business cycles exhibit *duration dependence*: whether expansions and contractions become more likely to end as they age. Using hazard-based tests on the NBER chronology, Diebold and Rudebusch (1990) and Diebold, Rudebusch, and Sichel (1993) find positive duration dependence in prewar expansions and postwar contractions but none in postwar expansions, and Rudebusch (2016) confirms that postwar expansions do not end out of old age. Our model is consistent with this evidence by construction: the probability of leaving the good regime is constant, so expansions do not die of old age. But older expansions leads to a severe downturn in intensive margin: a longer expansion leaves a denser inherited network, so the contraction that ends it is deeper (Corollary 4).

In this section, we do not re-estimate the aggregate plucking property in the data or target the one-sided predictability moment. Figure 7 instead shows that this moment emerges as an untargeted outcome of the simulated equilibrium. We then turn to the model's distinctive cross-sectional implication. We focus on sectoral employment dynamics, construct a peak-to-trough "drop" measure that maps naturally to the model's state variable (network intensity), and test whether industries with higher intermediate-input intensity are more exposed to downside risk, as implied by the theory.

## 6.2 Evidence for plucking through the networks

This section revisits the empirical "plucking" property in the cross-industry employment dynamics of U.S. manufacturing sectors. Our goal is twofold. First, we construct a recession-severity measure at the industry level that can be directly related to our model's state variable capturing network intensity. Second, we test the central implication of the model: more network-intensive industry-years exhibit deeper peak-to-trough contractions.

**Data** We use the NBER–CES Manufacturing Industry Database, which provides annual measures of industry employment and cost components at detailed NAICS levels. The sample spans 1958–2018, with industry-year observations indexed by NAICS code and calendar year.

**Identifying peaks and troughs** For each industry, we extract the cyclical component of employment using an HP filter for annual data (smoothing parameter $\lambda = 6.25$). Let $\widetilde{e}_{i,t}$ denote the HP cycle of employment for industry $i$ in year $t$. We identify a local peak as a year in which the cycle satisfies $\widetilde{e}_{i,t} > \widetilde{e}_{i,t-1}$ and $\widetilde{e}_{i,t} \geq \widetilde{e}_{i,t+1}$, and a local trough analogously as $\widetilde{e}_{i,t} < \widetilde{e}_{i,t-1}$ and $\widetilde{e}_{i,t} \leq \widetilde{e}_{i,t+1}$. For each peak, we then match it to the next trough in the same industry to form a peak–to–trough episode.

**Plucking depth** Given a matched peak year $t_{i,k}^{\text{peak}}$ and subsequent trough year $t_{i,k}^{\text{trough}}$ for episode $k$ in industry $i$, we define the contraction depth ("drop") as

$$\text{Drop}^{\text{emp}}{}_{i,k} \equiv \log\left(\text{emp}_{i,t_{i,k}^{\text{peak}}}\right) - \log\left(\text{emp}_{i,t_{i,k}^{\text{trough}}}\right).$$

This statistic is the industry-level counterpart of the peak-to-trough measures used in the empirical plucking literature, and it will be the main outcome variable in our network-based validation below.

We now test the central mechanism implied by the model: industries that operate with a more intermediate-input-intensive production structure are more vulnerable to downturns, exhibiting deeper peak-to-trough employment contractions. Since the NBER–CES data are not input–output matrices, we proxy network intensity using a cost-based intermediate-input measure.

**Network intensity proxy: material share** We construct an industry-year material share as

$$\text{material_share1}_{i,t} \equiv \frac{\text{matcost}_{i,t}}{\text{vship}_{i,t}},$$

where matcost is the cost of materials and vship is the value of shipments. In the model, the intermediate-input share is strictly increasing in effective inherited network intensity, $(1-\delta)\alpha$. The material share therefore provides a monotone empirical proxy for network exposure. [11]

**Procyclicality of network intensity** Before turning to the cross-sectional test, we verify a direct implication of the model: network intensity is a slow-moving state that is built up in expansions and shed gradually, not instantaneously, once a downturn begins (Corollary 2). Figure 9 traces the material share around each industry's own employment peaks and troughs, dated exactly as in the regressions below. The share does not fall in the peak year itself; it declines over the following three years, by 0.18, 0.52 and 0.62 percentage points, and only then begins to recover. Around troughs, the share keeps falling into the trough and is flat afterwards. Network intensity therefore lags the employment cycle, as the predetermined, slowly adjusting state of the model does, and the material share at the peak measures the network the industry carries into the contraction.

**Baseline specification** We estimate the relationship between contraction depth and predetermined network intensity using the following regression:

$$\text{Drop}^{Y}_{i,k} = \beta\, \text{material_share}_{i,t^{\text{peak}}_{i,k}} + \gamma X_{i,t^{\text{peak}}_{i,k}} + \mu_i + \lambda_{t^{\text{peak}}_{i,k}} + \varepsilon_{i,k},$$

where $Y$ is one of three outcome variables: employment (emp), total payroll (pay), or output (vship). The key regressor is the material share measured at the peak year, proxying for the industry's network exposure at the onset of the contraction. $\mu_i$ are industry fixed effects and $\lambda_{t^{\text{peak}}}$ are peak-year fixed effects. Standard errors are clustered at the industry level. The inclusion of peak-year fixed effects absorbs aggregate conditions at the onset of the contraction, so identification comes from cross-industry variation in material share within a given peak year. The control vector $X_{i,t^{\text{peak}}_{i,k}}$ includes the log capital stock at the peak year.

[11] Because matcost already includes purchased energy, the baseline measure includes energy costs. As a robustness check, we instead use the non-energy material share $\text{material_share2}_{i,t} \equiv \frac{\text{matcost}_{i,t} - \text{energy}_{i,t}}{\text{vship}_{i,t}}$. The results are qualitatively similar (Online Appendix E).

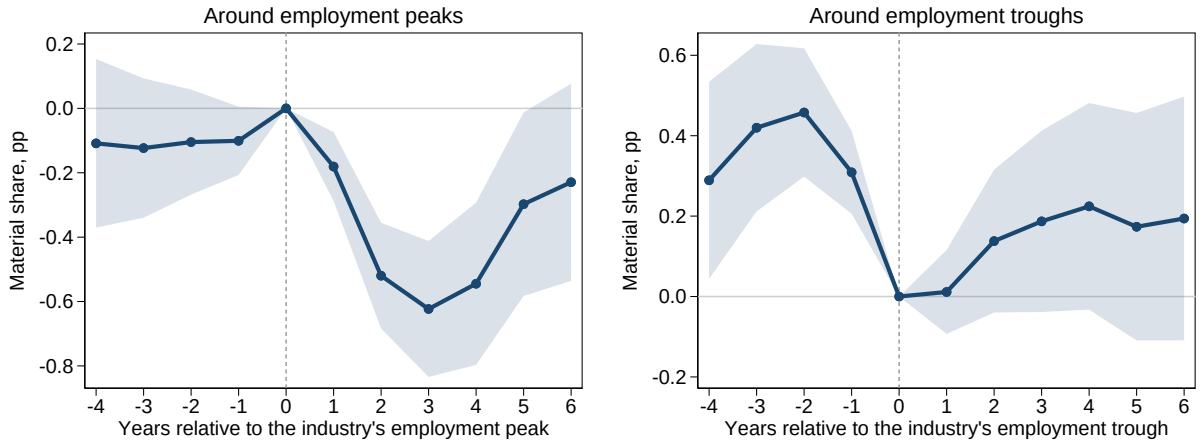



Figure 9: Material share around industry employment peaks and troughs

Notes: NBER–CES, 364 industries, 1958–2018. Material share = matcost/vship, in percentage points, relative to the turning-point year. Peaks and troughs are adjacent local extrema of the HP(6.25)-filtered level of industry employment, as in Table 5. Estimates come from a distributed-lag regression of the annual change in the share on leads and lags $(-6, \ldots, 6)$ of the turning-point indicator with industry fixed effects, so that overlapping event windows are attributed jointly; the path is the cumulative sum of the coefficients. Shaded areas are 95 percent confidence intervals, with standard errors clustered by industry.

**Results** Columns (1)–(3) of Table 5 report the estimates, with material share and boom duration entered jointly. The coefficient on material share is positive and significant for all three outcomes: industries with higher material intensity at the peak experience deeper peak-to-trough contractions in employment, payroll and gross output, holding constant how long the preceding expansion lasted. A 10-percentage-point higher material share implies a 1.2% deeper employment contraction and a 1.4% deeper contraction in gross output. Boom duration enters independently with the same sign for every outcome. Online Appendix E reports each regressor separately and the non-energy material share; the material-share coefficient is larger when entered alone (0.13 for employment, 0.17 for gross output).

Columns (4)–(6) apply the same episode regressions to a simulated path from the calibrated model. The coefficients on material share are positive for all three outcomes and survive the boom-duration control: the calibrated model reproduces the association observed in the data.

Table 5: Material share and peak-to-trough drop

| | Data | | | Model | | |
|---|---|---|---|---|---|---|
| $\text{Drop}^Y$, Y= | (1) emp | (2) pay | (3) vship | (4) emp | (5) pay | (6) vship |
| Material Cost Share | 0.120** | 0.123** | 0.141** | 0.096*** | 0.192*** | 0.273*** |
| | (0.043) | (0.045) | (0.054) | (0.002) | (0.005) | (0.023) |
| Booming Duration | 0.014*** | 0.011*** | 0.014*** | 0.0005*** | 0.0010*** | 0.0353*** |
| | (0.002) | (0.002) | (0.003) | (0.0001) | (0.0003) | (0.0015) |
| Industry FE | Yes | Yes | Yes | | | |
| Peak year FE | Yes | Yes | Yes | | | |
| R-squared | 0.365 | 0.367 | 0.326 | 0.397 | 0.397 | 0.287 |
| N | 5518 | 5518 | 5518 | 2317 | 2317 | 2317 |

*Notes:* The dependent variable is the log peak-to-trough drop in employment, payroll, or the value of shipments (vship). Peaks and troughs are adjacent local extrema of the HP-filtered level of employment, and material share is measured at the peak. Booming duration is the number of years from the previous trough to the peak. Columns (1)–(3) include industry and peak-year fixed effects and peak capital stock; standard errors are clustered by industry. Columns (4)–(6) apply the same local-extrema construction to the calibrated model's ergodic sample and report heteroskedasticity-robust standard errors. Online Appendix E reports each regressor separately. Standard errors are in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

# 7 Concluding remarks

This paper proposes a theory of endogenous plucking in which business-cycle asymmetry arises not from skewed shocks, but from the endogenous evolution of the production structure. During expansions, firms accumulate network linkages that raise productivity, yet these same linkages amplify contractions when aggregate productivity falls—generating duration dependence and negative skewness in output growth consistent with the data.

Our normative analysis uncovers a Plucking Paradox: the retail markup makes the calibrated decentralized economy, in a welfare sense, "too safe" —featuring shallower supply chains and milder recessions, but also lower average consumption. The Pareto-efficient allocation entails deeper networks, higher output, and sharper downturns. Correcting the markup therefore requires tolerating—and in good times, even encouraging—a particular form of structural fragility.

A natural extension is to study interactions with monetary policy. In our framework, the economy's sensitivity to shocks is not fixed but rises endogenously with the duration of an expansion as network intensity accumulates. This means the central bank faces a

stabilization problem that evolves over the cycle: the same interest rate intervention has different effects depending on the inherited network state. Effective policy must therefore manage booms not only for inflation and output, but also for the evolving fragility of the production structure.

# References


Acemoglu, Daron and Pablo D. Azar. 2020. "Endogenous Production Networks." *Econometrica* 88 (1):33–82. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA15899.

Acemoglu, Daron, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. 2012. "The Network Origins of Aggregate Fluctuations." *Econometrica* 80 (5):1977–2016. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA9623.

Acemoglu, Daron, Asuman Ozdaglar, and Alireza Tahbaz-Salehi. 2017. "Microeconomic Origins of Macroeconomic Tail Risks." *American Economic Review* 107 (1):54–108. URL https://www.aeaweb.org/articles?id=10.1257/aer.20151086.

Antràs, Pol, Davin Chor, Thibault Fally, and Russell Hillberry. 2012. "Measuring the Upstreamness of Production and Trade Flows." *American Economic Review* 102 (3):412–16. URL https://www.aeaweb.org/articles?id=10.1257/aer.102.3.412.

Argente, David, Sara Moreira, Ezra Oberfield, and Venky Venkateswaran. 2025. "Scalable Expertise: How Standardization Drives Scale and Scope." NBER Working Papers 34160, National Bureau of Economic Research, Inc. URL https://ideas.repec.org/p/nbr/nberwo/34160.html.

Baqaee, David Rezza and Emmanuel Farhi. 2019. "The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem." *Econometrica* 87 (4):1155–1203. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA15202.

Basu, Susanto. 1995. "Intermediate Goods and Business Cycles: Implications for Productivity and Welfare." *The American Economic Review* 85 (3):512–531. URL http://www.jstor.org/stable/2118185.

Becker, Randy A., Wayne B. Gray, and Jordan Marvakov. 2021. "NBER–CES Manufacturing Industry Database (1958–2018, version 2021a)." National

Bureau of Economic Research. URL https://www.nber.org/research/data/nber-ces-manufacturing-industry-database.

Burdett, Kenneth, Shouyong Shi, and Randall Wright. 2001. "Pricing and Matching with Frictions." *Journal of Political Economy* 109 (5):1060–1085. URL https://doi.org/10.1086/322835.

Butters, Gerard R. 1977. "Equilibrium Distributions of Sales and Advertising Prices." *The Review of Economic Studies* 44 (3):465–491. URL https://doi.org/10.2307/2296902.

Capponi, Agostino, Chuan Du, and Joseph E Stiglitz. 2024. "Are Supply Networks Efficiently Resilient?" Working Paper 32221, National Bureau of Economic Research. URL http://www.nber.org/papers/w32221.

Carvalho, Vasco M, Makoto Nirei, Yukiko U Saito, and Alireza Tahbaz-Salehi. 2020. "Supply Chain Disruptions: Evidence from the Great East Japan Earthquake*." *The Quarterly Journal of Economics* 136 (2):1255–1321. URL https://doi.org/10.1093/qje/qjaa044.

Demir, Banu, Ana Cecília Fieler, Daniel Yi Xu, and Kelly Kaili Yang. 2024. "O-Ring Production Networks." *Journal of Political Economy* 132 (1):200–247. URL https://doi.org/10.1086/725703.

Diebold, Francis X. and Glenn D. Rudebusch. 1990. "A Nonparametric Investigation of Duration Dependence in the American Business Cycle." *Journal of Political Economy* 98 (3):596–616.

Diebold, Francis X., Glenn D. Rudebusch, and Daniel E. Sichel. 1993. "Further Evidence on Business-Cycle Duration Dependence." In *Business Cycles, Indicators and Forecasting*, edited by James H. Stock and Mark W. Watson. Chicago: University of Chicago Press, 255–284.

Dupraz, Stéphane, Emi Nakamura, and Jón Steinsson. 2025. "A plucking model of business cycles." *Journal of Monetary Economics* 152:103766.

Ferrari, Alessandro and Lorenzo Pesaresi. 2026. "Specialization, Complexity, and Resilience in Supply Chains." Working paper.

Friedman, Milton. 1964. "Monetary Studies of the National Bureau." In *The National Bureau Enters Its 45th Year*, 44th Annual Report. New York: National Bureau of Economic Research, 7–25.

Ghassibe, Mishel. 2021. "Endogenous Production Networks and Non-Linear Monetary Transmission." Working paper.

Ghassibe, Mishel and Anton Nakov. 2025. "Business Cycles with Pricing Cascades." Working paper.

Hall, Robert E. 1979. "A Theory of the Natural Unemployment Rate and the Duration of Employment." *Journal of Monetary Economics* 5 (2):153–169. URL https://web.stanford.edu/~rehall/Theory_Natural_Unemployment.pdf.

Hall, Robert E. and Marianna Kudlyak. 2022a. "The inexorable recoveries of unemployment." *Journal of Monetary Economics* 131:15–25. URL https://www.sciencedirect.com/science/article/pii/S0304393222000939.

———. 2022b. "Why Has the US Economy Recovered So Consistently from Every Recession in the Past 70 Years?" *NBER Macroeconomics Annual* 36:1–55. URL https://doi.org/10.1086/718588.

Hartley, Jonathan S. 2021. "Friedman's plucking model: New international evidence from Maddison Project data." *Economics Letters* 199:109724.

Hulten, Charles R. 1978. "Growth Accounting with Intermediate Inputs." *The Review of Economic Studies* 45 (3):511–518. URL https://ideas.repec.org/a/oup/restud/v45y1978i3p511-518..html.

Hyun, Jay, Ryan Kim, and Byoungchan Lee. 2024. "BUSINESS CYCLES WITH CYCLICAL RETURNS TO SCALE." *International Economic Review* 65 (1):253–282. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/iere.12656.

Keane, Michael and Richard Rogerson. 2012. "Micro and Macro Labor Supply Elasticities: A Reassessment of Conventional Wisdom." *Journal of Economic Literature* 50 (2):464–476.

Kim, Chang-Jin and Charles R. Nelson. 1999. "Friedman's Plucking Model of Business Fluctuations: Tests and Estimates of Permanent and Transitory Components." *Journal of Money, Credit and Banking* 31 (3):317–334.

Kim, Chang-Jin and Jeremy Piger. 2002. "Common Stochastic Trends, Common Cycles, and Asymmetry in Economic Fluctuations." *Journal of Monetary Economics* 49 (6):1189–1211.

Kopytov, Alexandr, Bineet Mishra, Kristoffer Nimark, and Mathieu Taschereau-Dumouchel. 2024. "Endogenous Production Networks under Supply Chain Uncertainty." *Econometrica* 92 (5):1621–1659. URL https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA20629.

Kopytov, Alexandr, Mathieu Taschereau-Dumouchel, and Zebang Xu. 2025. "The Origin of Risk." Working paper.

———. 2026. "Endogenous Returns to Scale." Working paper.

Krusell, Per and Anthony A. Smith, Jr. 1998. "Income and Wealth Heterogeneity in the Macroeconomy." *Journal of Political Economy* 106 (5):867–896. URL https://www.journals.uchicago.edu/doi/10.1086/250034.

Lashkari, Danial, Arthur Bauer, and Jocelyn Boussard. 2024. "Information Technology and Returns to Scale." *American Economic Review* 114 (6):1769–1815. URL https://www.aeaweb.org/articles?id=10.1257/aer.20220522.

Lee, Hanbaek. 2026a. "Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics." Working paper.

———. 2026b. "A Global Dynamic Nonlinear Solution Framework and the Repeated Transition Method." Working paper.

Neftçi, Salih N. 1984. "Are Economic Time Series Asymmetric over the Business Cycle?" *Journal of Political Economy* 92 (2):307–328. URL https://doi.org/10.1086/261226.

Oberfield, Ezra. 2018. "A Theory of Input-Output Architecture." *Econometrica* 86 (2):559–589.

Petrongolo, Barbara and Christopher A. Pissarides. 2001. "Looking into the Black Box: A Survey of the Matching Function." *Journal of Economic Literature* 39 (2):390–431. URL https://www.aeaweb.org/articles?id=10.1257/jel.39.2.390.

Pissarides, Christopher A. 1979. "Job Matchings with State Employment Agencies and Random Search." *The Economic Journal* 89 (356):818–833. URL https://doi.org/10.2307/2231501.

Rudebusch, Glenn D. 2016. "Will the Economic Recovery Die of Old Age?" FRBSF Economic Letter 2016-03, Federal Reserve Bank of San Francisco. URL https://www.frbsf.org/research-and-insights/publications/economic-letter/2016/02/will-economic-recovery-die-of-old-age/.

Stevens, Margaret. 2007. "New Microfoundations for the Aggregate Matching Function, with Empirical and Theoretical Implications." *International Economic Review* 48 (3):847–868. URL https://doi.org/10.1111/j.1468-2354.2007.00447.x.

Taschereau-Dumouchel, Mathieu. 2025. "Cascades and Fluctuations in an Economy with an Endogenous Production Network." *The Review of Economic Studies* :rdaf036.

Topkis, Donald M. 1978. "Minimizing a Submodular Function on a Lattice." *Operations Research* 26 (2):305–321. URL http://www.jstor.org/stable/169636.